

Ev-Trust: An Evolutionarily Stable Trust Mechanism for Decentralized LLM-Based Multi-Agent Service Economies

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ABSTRACT

Decentralized LLM-based multi-agent service economies face three vulnerabilities that undermine traditional trust mechanisms: reduced cost of fraud, difficulty in evaluating service quality, and instability of service content. These compounding vulnerabilities can trigger population-level trust collapse and the proliferation of short-sighted strategies. We propose *Ev-Trust*, an evolutionarily stable trust mechanism that addresses these vulnerabilities through three targeted designs: a cross-validation gate leveraging requestor semantic comprehension to assess response validity, a variance-standardized drift measure filtering endogenous stochasticity from genuine behavioral anomalies, and an embedding of trust signals into the expected revenue function that converts trustworthiness into an evolutionary survival advantage. Based on replicator dynamics with a noisy best response micro-foundation, we prove the asymptotic stability of cooperative evolutionarily stable strategies and derive explicit threshold conditions for maintaining cooperative equilibria. We evaluate *Ev-Trust* through 100-round simulations with at least 100 heterogeneous LLM-driven agents covering seven behavioral types. The experiments are conducted on TruthfulQA and TriviaQA, two factual question-answering benchmarks. Compared to baselines based on transitive trust aggregation, reinforcement-learning reputation, and pure evolutionary imitation, *Ev-Trust* reduces malicious agent participation by approximately 60%, suppresses the fraudulent service rate by approximately 50%, and maintains stable trust differentiation under a 30% adversarial mutation. These results demonstrate that coupling semantic trust evaluation with evolutionary incentives provides a principled foundation for securing cooperation in decentralized LLM-based multi-agent systems.

1. INTRODUCTION

Recently, the proliferation of API-level large language models (LLMs) has catalyzed the deployment of LLM-based agents across diverse applications, including data analysis (Chen et al., 2026), code refactoring (Wu et al., 2024), and automated planning (Li et al., 2023). Nevertheless, the conventional centralized agent paradigm imposes severe structural limitations. Vendor lock-in restricts service capabilities, platform silos constrain multi-agent collaboration, and centralized vulnerabilities expose downstream users to cascading failures. Driven by these bottlenecks, AI agents are evolving from isolated tools into autonomous entities within decentralized service markets. Platforms such as Olas (Minarsch et al., 2023; Jung and Lee, 2026) now enable autonomous economic agents to execute cross-domain tasks without central intermediaries, while Bittensor (Rao et al., 2020) implements a peer-to-peer LLM network safeguarded by game-theoretic trust mechanisms. Ultimately, as decentralized service markets emerge as a transformative architecture for multi-agent systems, establishing robust security and operational stability has become the field’s most critical imperative.

However, this decentralized model destabilizes the behavioral foundations upon which agents form reliable expectations about their interaction partners, and it also introduces risks uniquely amplified from LLMs: **(i) Reduced cost of fraud.** Compared to the significant marginal cost of fraudulent outputs in traditional service markets, LLM-based agents can generate responses that are factually incorrect yet stylistically professional at virtually the same cost

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(O’Gara, 2023; Spitale et al., 2023; Han et al., 2025). Extremely low fraud costs incentivize fraudulent strategies. **(ii) Difficulty in evaluating service quality.** Traditional services have objective quality standards, whereas LLM outputs consist of high-dimensional semantic content, lacking precise evaluation benchmarks (Xu et al., 2023; Ji et al., 2023). The absence of objective quality benchmarks delays the detection of risky behavior. **(iii) Instability of service content.** Even when the same agent processes the same input, it may produce different outputs due to the inherent instability of LLMs. This inherent output randomness makes it difficult to distinguish between benign randomness and malicious strategy shifts (Haque et al., 2025; Ouyang et al., 2023). As these vulnerabilities compound, the basis for reliably assessing partner behavior degrades, trust relationships between agents collapse, and short-sighted strategies proliferate unchecked.

We further distinguish our problem scope from adjacent directions. Communication security and identity management address external threats such as message tampering and Sybil attacks through authentication and filtering (Yu et al., 2024), but this protects the reliability of the channel rather than the authenticity of the content: an authenticated agent can generate semantically fraudulent responses through a fully compliant secure channel. Reinforcement learning prioritizes individual policy optimization (Ba et al., 2024), which operates at the individual level and does not address population-level evolutionary stability. Traditional multi-agent trust mechanisms typically rely on static reputation models or bounded strategy sets (Kamvar et al., 2003; Jurca and Faltings, 2007), which assume objectively verifiable service quality and exogenous behavioral variability, premises that break down in LLM service markets. Existing methods each address a specific class of threats, but none simultaneously confronts the threefold vulnerability of the LLM service market identified above. Restoring a reliable basis for evaluating partner behavior and suppressing strategic fraud in decentralized LLM-based service economies without central authority remains an open challenge.

To address these challenges, we propose *Ev-Trust*, an evolutionarily stable trust mechanism for decentralized LLM-based multi-agent service economies. To the best of our knowledge, this is the first mechanism that couples LLM semantic trust evaluation with evolutionary stable dynamics in a fully decentralized setting.

1.1. Research Objectives

This study addresses the threefold vulnerability of decentralized LLM-based service markets, namely near-zero fraud cost, semantic quality verification difficulty, and endogenous output stochasticity, through a trust mechanism that couples semantic trust evaluation with evolutionary game-theoretic dynamics. The primary research objectives are as follows:

- To design LLM-specific trust metrics, a cross-validation gate for quality assessment and a variance-standardized drift measure separating endogenous stochasticity from malicious anomalies, both operating on locally observable signals.
- To develop a trust-revenue coupling that embeds trust signals into the agent’s expected revenue function, converting trustworthiness into a decisive survival advantage, suppressing fraudulent strategies.
- To provide rigorous theoretical guarantees based on Replicator Dynamics (Taylor and Jonker, 1978) with a noisy best response micro-foundation (McKelvey and Palfrey, 1995), deriving boundary conditions and proving the asymptotic stability of the cooperative ESS (Maynard Smith, 1976).
- To empirically validate the mechanism through experiments with heterogeneous LLM-driven agents, benchmarking against baselines from transitive trust aggregation, reinforcement-learning reputation, and evolutionary imitation paradigms.

By pursuing these objectives, this work aims to re-establish a reliable basis for evaluating partner behavior in decentralized LLM-based service markets without central authority, and to demonstrate that coupling semantic trust evaluation with evolutionary incentives drives autonomous agent populations toward robust cooperation.

1.2. Contributions

In summary, the main contributions of this paper are as follows:

- We propose *Ev-Trust*, the first decentralized trust mechanism coupling LLM semantic trust evaluation with evolutionary stable dynamics, converting trustworthiness into a decisive survival advantage.

- We design two semantic trust metrics for LLM-specific vulnerabilities, neutralizing the deceptive advantage of plausible yet incorrect outputs and statistically disentangling benign stochasticity from malicious strategy shifts.
- We prove that when the long-term trust-induced gain exceeds the short-term fraud incentive, the cooperative equilibrium is evolutionarily stable and globally attracting.
- We conduct extensive experiments with heterogeneous LLM-driven agents, showing that *Ev-Trust* eliminates malicious strategies, maintains correct economic incentive hierarchies, and remains resilient under mutant invasion.

2. Related Work

2.1. Vulnerability Analysis of Decentralized LLM Services

LLM-based service markets exhibit a distinct vulnerability profile that sets them apart from traditional service economies. The near-zero cost of generating fraudulent content, the absence of objective quality benchmarks for semantic outputs, and the inherent stochasticity of autoregressive generation collectively create a compounding risk structure. **(i)** In the dimension of fraud behavior, LLM-based agents have been shown to spontaneously develop deceptive strategies in competitive settings (Aitchison et al., 2021; O’Gara, 2023), exploiting low-cost generation to maximize short-term profit (Zhao et al., 2023). Motwani et al. (2024) revealed that deceptive strategies propagate through steganographic collusion channels, while Ju et al. (2024); Xiong et al. (2025) showed that manipulated knowledge amplifies the impact of individual fraud at the collective level. Pan et al. (2023) further found that LLM agents exhibit Machiavellian behavior even in cooperative scenarios. **(ii)** Regarding service quality verification, Xu et al. (2023) showed that LLM agents exhibit nuanced strategic behavior that resists binary classification, and Haque et al. (2025) documented that semantically fluent but factually flawed outputs are pervasive and difficult to detect through surface-level metrics. Standard automated evaluation metrics correlate poorly with human judgments of factual accuracy (Maynez et al., 2020), and Lin et al. (2022) demonstrated that even state-of-the-art LLMs confidently generate plausible but incorrect answers on questions involving common misconceptions. **(iii)** In the dimension of output stochasticity, identical inputs yield qualitatively different outputs across invocations even under fixed decoding parameters (Ouyang et al., 2023), a fundamental property of autoregressive generation rather than a correctable defect (Holtzman et al., 2020), directly confounding the use of output consistency as a reliability signal.

Existing defense approaches each address a subset of these challenges. Centralized supervision (Xiang et al., 2024; Zhang et al., 2024b) and agent-level safety frameworks (Hua et al., 2024; Ren et al., 2024) depend on high-authority arbitration or privileged access to agent internals, assumptions incompatible with decentralized markets. Communication security and identity management approaches (Lee and Tiwari, 2024; Gu et al., 2024; Yu et al., 2024) protect channel integrity but cannot detect authenticated agents generating semantically fraudulent content through compliant channels. Reinforcement learning approaches, including recent work such as LR2 (Ren et al., 2025) that introduces decentralized reputation reward shaping, optimize individual agent behavior but provide no formal guarantees on population-level stability (Ba et al., 2024). None of the above methods simultaneously confronts all three LLM-specific vulnerabilities in a fully decentralized setting.

2.2. Trust Mechanisms in Multi-Agent Systems

Trust mechanisms have a long research history in distributed and multi-agent systems. Early reputation systems for electronic markets (Resnick et al., 2006; Cabral and Hortacsu, 2010) demonstrated that historical feedback can regulate seller behavior in centralized platforms. Kamvar et al. (2003) proposed the EigenTrust algorithm for peer-to-peer networks via transitive propagation. Josang and Ismail (2002) introduced the Beta Reputation System using Bayesian inference over binary outcomes, extended by Jøsang and Haller (2007) to multi-level outcomes through Dirichlet distributions, while Jurca and Faltings (2007) addressed incentive compatibility through payment rules. These foundational works established key paradigms for decentralized trust computation, but assume objectively verifiable service quality and exogenous behavioral variability, both of which break down in LLM service markets.

Recent research has begun to account for the characteristics of LLM-based agents. Wang et al. (2026) proposed a cognitive trust framework for multi-dimensional behavioral profiling, while Gao et al. (2025) introduced trust-aware architectures for dynamic coordination. Wang et al. (2023) developed benchmark suites for evaluating LLM trustworthiness, and Jiang et al. (2026) proposed reputation mechanisms for AI-generated content. Yet these approaches generally require centralized evaluation authorities, privileged access to agent internals, or substantial intervention in

agent decision-making, rendering them incompatible with open, decentralized service economies. This motivates the need for a trust framework that operates with only locally observable signals, a requirement that naturally aligns with evolutionary approaches to collective regulation.

Table 1
Agent Types in the Decentralized LLM-Based Service Market.

Type	Role Description	Vulnerability	Rational	Evolvable
R_n	Rational participant. Maximize expected utility and engages in fair mutual evaluation	Service quality lacks objective verification benchmarks (ii)	✓	✓
R_{m1}	Trust saboteur. Propagates false recommendations to neighbors	Additional Threats from Collusive Attacks	✗	✓
R_{m2}	Random participant. Behaves randomly without strategic intent	Distinguishing between endogenous instability and strategic change (iii)	✗	✗
P_n	Rational participant. Delivers accurate responses to maximize long-term expected revenue	Output quality fluctuates due to endogenous model stochasticity (iii)	✓	✓
P_{m1}	Short-sighted fraudster. Generates responses that appear relevant but contain factual errors	Fraud enables plausible yet incorrect outputs (i)	✗	✓
P_{m2}	Cost minimizer. Delivers superficial responses with minimal computational effort	Low-cost low-quality service is difficult to assess (i, ii)	✗	✓
P_{m3}	Strategic opportunist. Provides high-quality service to accumulate trust, then exploits earned reputation	Zero switching cost makes strategy shifts indistinguishable from normal variance (iii)	✗	✓

2.3. Evolutionary Game Theoretic Approaches

Game theory provides a foundational framework for modeling strategic interdependence in multi-agent systems (Von Neumann and Morgenstern, 1944). Classical evolutionary game theory extended this to population-level dynamics through Evolutionarily Stable Strategies (Maynard Smith, 1976) and Replicator Dynamics (Taylor and Jonker, 1978). Hofbauer and Sigmund (2003) formalized the connection between individual learning rules and replicator equations, while Sandholm (2010) comprehensively treated how diverse micro-level revision protocols give rise to aggregate dynamics. These foundations have been widely applied to explain cooperation in biological and social systems (Nowak, 2006).

Evolutionary game-theoretic approaches have proved crucial for regulating collective behavior in multi-agent systems. Bloembergen et al. (2015) surveyed the interplay between evolutionary dynamics and multi-agent learning, and Han (2022) demonstrated that emergent cooperative behaviors can be predicted through evolutionary mechanisms. Sarkadi and Lewis (2024) modeled the evolution of dishonesty in agent societies, revealing that deceptive strategies become dominant without appropriate countermeasures. Recent studies have further extended these mechanisms to social governance: Yang et al. (2025) developed a multi-party evolutionary model for collaborative governance of malicious online behaviors, and Xu et al. (2025) analyzed strategy adaptation and trust dynamics in data-sharing ecosystems. Nowak and Sigmund (1998) and Ohtsuki et al. (2006) respectively studied reputation-based partner selection and network topology effects on evolutionary stability of cooperation.

More recently, researchers have extended game-theoretic approaches to LLM-based agents. Akata et al. (2025) and Guo (2023) revealed that generative agents exhibit human-like strategic reasoning in repeated games, while Brookins and Edelman (2023) investigated in-context strategy adaptation and Ren et al. (2024) examined the emergence of social norms in LLM populations. However, most current work relies on static Nash equilibria, analyzing individual reasoning rather than long-term evolutionary stability (Guo et al., 2024). No existing work integrates evolutionary stability analysis with semantic trust evaluation tailored to LLM-specific vulnerabilities. Our work bridges this gap by integrating LLM-specific semantic trust evaluation into an evolutionary game-theoretic framework, enabling trust to function not merely as a passive record but as an active evolutionary force that drives decentralized agent populations toward stable cooperation.

3. PROBLEM FORMULATION

In this section, we model the transaction scenario for a decentralized multi-agent service within a formal game-theoretic framework, specifying the types of agents, the strategy space, the interaction rules, and the payoff structure.

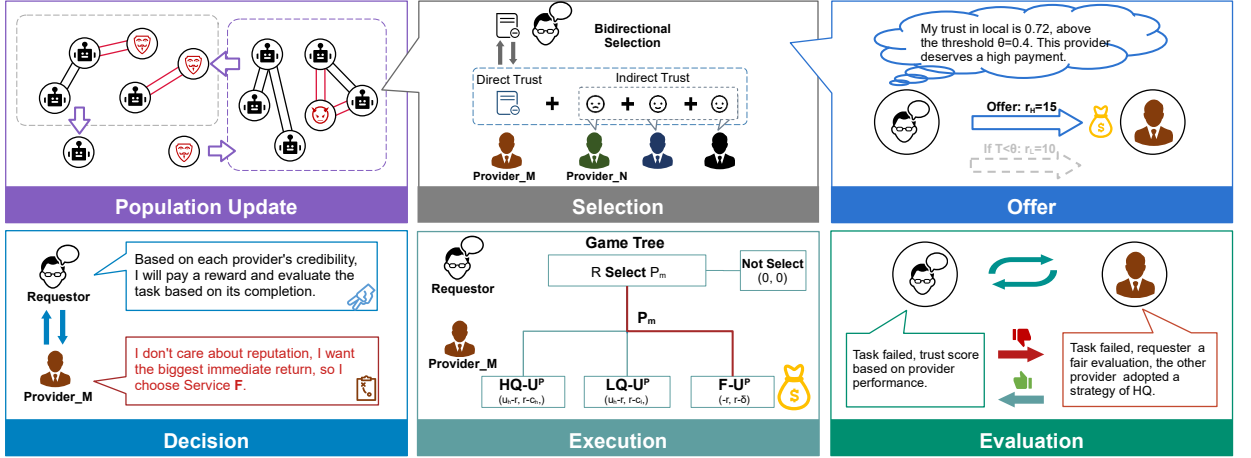


Figure 1: Illustrative example of a single-round interaction following the six-stage workflow. The requestor selects a provider via softmax matching, offers trust-differentiated payment, and both parties update local trust after service execution.

3.1. Scenario Setting

We consider a decentralized multi-agent service market that operates in discrete time steps $t \in \{1, \dots, N\}$, populated by agents acting as Requestors (R) or Providers (P). Guided by evolutionary game rules, agents adaptively update their behavioral strategies based on expected revenue. Specifically, we model evolutionary game interactions as exchanges of semantic messages. Each interaction involves a **Request Vector** $\mathbf{v}_{req}$ (task description) and a **Response Vector** $\mathbf{v}_{resp}$ (service output). This semantic layer enables agents to evaluate quality and consistency beyond binary outcomes. Formally, let $\mathcal{A}_t = \mathcal{R}_t \cup \mathcal{P}_t$ denote the active population at time t . The sets evolve dynamically as:

$$\mathcal{A}_t = (\mathcal{A}_{t-1} \cup \Delta \mathcal{A}_t^+) \setminus \Delta \mathcal{A}_t^-, \quad \forall \mathcal{A} \in \{\mathcal{R}, \mathcal{P}\}, \quad (1)$$

where $\Delta \mathcal{A}_t^\pm$ represent entering/exiting agents. To strictly quantify the LLM-specific risks within the service market, we introduce heterogeneous agent types that reflect distinct threat modalities unique to or amplified by LLM-based services. Table 1 details the behavioral logic, LLM-specific threat source, and evolutionary capability of each type.

To formalize the decision-making process and capture the strategic trade-offs between service quality and operational cost, we define the provider's strategy space as $S^P = \{HQ, LQ, F, D\}$. These four active strategies correspond to a natural categorization in service economics: qualified delivery (HQ), substandard but honest delivery (LQ), and deceptive delivery (F). High-quality service (HQ) invests sufficient computational effort to produce accurate, complete responses, yielding long-term trust gains. Low-quality service (LQ) minimizes generation cost by producing brief, superficial responses that are partially relevant but insufficient. Fraudulent service (F) exploits the near-zero fraud cost of LLMs to deliver responses that may appear relevant yet fail to satisfy the task requirements, yielding high immediate profits at minimal generation cost. Denial of service (D) is a passive strategy in which the provider declines interaction based on low trust in the requestor.

This categorization reflects a natural partition of the service quality spectrum (qualified, substandard, and deceptive delivery) rather than a mechanism-specific design, and extends naturally to finer quality grades without modifying the trust metrics or evolutionary analysis framework. In LLM-based markets, this partition is especially consequential: the marginal generation cost varies minimally across quality levels, and the execution cost of F approaches zero, significantly amplifying the attractiveness of fraudulent strategies.

3.2. Game Workflow

We formalize the decision logic and revenue structure of the evolutionary game. The market operates with independent interactions, and agents act autonomously within the population. We define *trust* as the subjective probability that a target agent will engage in favorable behavior during future interactions, denoted by $T_t(A, B)$ for the trust agent A places in B at time t . Based on this metric, we introduce two decision boundaries: the *payment*

threshold $\theta \in (0, 1)$, which determines whether a requestor offers a high payment (r_H) or a base payment (r_L); and the interaction threshold $\tau \in (0, 1)$, which governs the provider's willingness to accept a request.

The game payoff structure is parameterized by execution costs c_h and c_l for HQ and LQ services, the fraud risk cost δ , and utility gains u_h and u_l . To ensure a well-formed game, these parameters satisfy three constraints: positive margins for honest providers ($r_H > c_h > c_l > 0$), requestor participation incentives ($u_h > r_H, u_l > r_L$), and a fraud-incentive condition ($c_h > \delta$) under which the provider's immediate margin from fraud exceeds that from honest service. This last condition is the source of short-term fraud incentives that the trust mechanism must counteract. The detailed parameter justification is provided in Appendix A.

Table 2

Game Revenue Matrix. Each cell (u^P, u^R) denotes the immediate revenue of the provider and requestor for a given strategy-payment combination.

P\R	Select, r_H	Select, r_L	NotSelect
HQ	$(r_H - c_h, u_h - r_H)$	$(r_L - c_h, u_h - r_L)$	(0,0)
LQ	$(r_H - c_l, u_l - r_H)$	$(r_L - c_l, u_l - r_L)$	(0,0)
F	$(r_H - \delta, -r_H)$	$(r_L - \delta, -r_L)$	(0,0)
D	(0,0)	(0,0)	(0,0)

The interaction process for each round proceeds through six stages. **(1) Population update (Update):** new agents with random types are admitted into the market, maintaining population dynamics. **(2) Trust-based provider selection (Selection):** each requestor selects a provider using softmax probability weighted by composite trust scores, ensuring that higher-trust providers are more likely to be chosen while preserving exploration opportunities for newer agents. **(3) Trust-differentiated payment offer (Offer):** the requestor determines the payment level based on the trust payment threshold θ , offering r_H to providers above the threshold and r_L otherwise. **(4) Provider acceptance and strategy decision (Decision):** the selected provider evaluates the requestor's trustworthiness against the interaction threshold τ and decides whether to accept; upon acceptance, the provider selects a service strategy from $\{HQ, LQ, F\}$ based on expected revenue. **(5) Service execution (Execution):** the provider generates a response under the chosen strategy, and both parties receive immediate revenue determined by the game matrix. **(6) Bilateral trust evaluation (Evaluation):** the requestor evaluates the response through semantic alignment S_{align} and behavioral drift D_{drift} , updating trust via Bayesian posterior inference; the provider evaluates payment fairness and updates trust via exponential moving average. The formal specification of each Stage is detailed as follow:

Stage 1: Population Update. Admit new agents into $\mathcal{R}_t, \mathcal{P}_t$.

Stage 2: Selection. Select provider j^* with probability $\propto \exp(\beta_s \cdot T_t(i, j))$; compute aggregated trust $T_t(i, j^*) \leftarrow \omega \cdot T_t^{direct}(i, j^*) + (1 - \omega) \cdot T_t^{indirect}(i, j^*)$.

Stage 3: Offer. Set payment $r_{ij^*} \leftarrow r_H$ if $T_t(i, j^*) > \theta$, else r_L .

Stage 4: Decision. If $T_t(j^*, i) < \tau$, provider denies interaction ($s_{j^*} \leftarrow D$). Otherwise, provider j^* selects strategy $s_{j^*} \in \{HQ, LQ, F\}$ based on expected revenue.

Stage 5: Execution. If $s_{j^*} \neq D$: provider j^* generates response $\mathbf{v}_{resp}$ under strategy s_{j^*} ; compute immediate revenue (u_t^R, u_t^P) :

$$\begin{cases} (u_h - r_{ij^*}, r_{ij^*} - c_h) & \text{if } s_{j^*} = HQ \\ (u_l - r_{ij^*}, r_{ij^*} - c_l) & \text{if } s_{j^*} = LQ \\ (-r_{ij^*}, r_{ij^*} - \delta) & \text{if } s_{j^*} = F \end{cases}$$

Stage 6: Evaluation. Requestor i evaluates $\mathbf{v}_{resp}$ via S_{align}, D_{drift} : Bayesian update $T_{t+1}(i, j^*)$. Provider j^* evaluates payment fairness: EMA update $T_{t+1}(j^*, i)$.

An illustrative example of the six-Stage interaction process is shown in Figure 1, where a requestor selects a provider through trust-based matching, offers a trust-differentiated payment, and both parties update trust through bilateral evaluation after service execution.

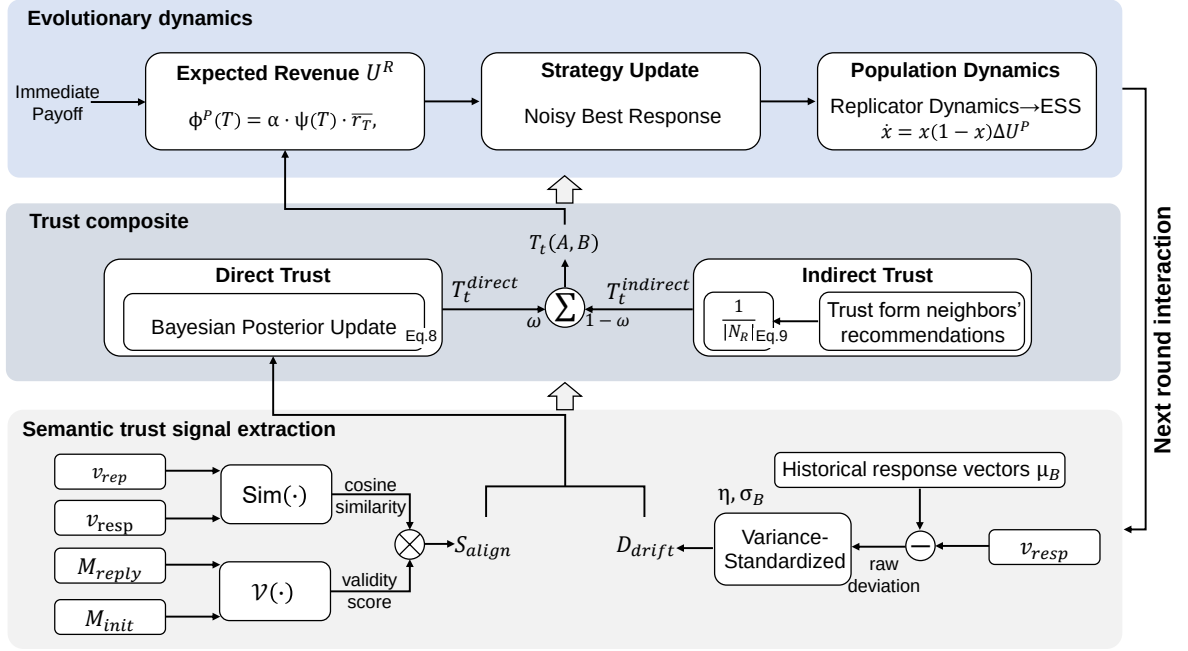


Figure 2: Architecture of *Ev-Trust*. Three layers: semantic trust signal extraction, trust composition via Bayesian update and indirect aggregation, and evolutionary dynamics coupling trust into expected revenue.

4. The Proposed Method

This section presents the core mechanism of *Ev-Trust*, an evolutionarily stable trust mechanism for decentralized LLM-based multi-agent service economies, whose overall architecture is illustrated in Figure 2. We first describe the metrics used to extract reliable signals from LLM interactions. Subsequently, we combine these metrics with the opinions of trusted neighbors to form direct and indirect trust, respectively, which together constitute the trust structure in the *Ev-Trust* method (Section 4.1). Second, we systematically describe how to embed the trust derived from these metrics into the expected revenue of agent games to drive strategy evolution, thereby realizing the complete *Ev-Trust* method (Section 4.2). Finally, we derive the conditions for the evolutionary stability of the *Ev-Trust* method based on replicator dynamics (Section 4.3).

4.1. Trust Composition and Metrics

The core of *Ev-Trust* is to transform observable semantic signals from LLM interactions into reliable trust measurements. The trust metric composition is designed to address the LLM-specific challenges: reduced cost of fraud, difficulty in evaluating service quality, and instability of service content.

The aggregate trust $T_t(A, B)$ of agent A toward agent B is a weighted combination of direct trust and indirect trust:

$$T_t(A, B) = \omega \cdot T_t^{direct}(A, B) + (1 - \omega) \cdot T_t^{indirect}(A, B), \quad (2)$$

where $\omega \in [0, 1]$ balances the reliance on personal history versus group recommendations. In practice, $\omega > 0.5$ ensures that direct trust dominates, with indirect trust primarily serving the cold-start scenario. Each agent stores and updates trust values locally without access to any global reputation registry, ensuring full compatibility with the decentralized architecture.

4.1.1. Semantic Trust Indicators

To ground the trust model in the generative behavior of LLM-based agents, we introduce two metrics derived from interaction messages. Let M_{init} and M_{reply} denote the raw text of the initiating message and the response, and let v_{init} and v_{reply} denote their corresponding embedding vectors.

Semantic Alignment: Measures the extent to which the response satisfactorily addresses the request. To counter fraud, where responses exhibit high surface-level relevance yet contain factual errors, we formulate $S_{align}^{(t)}$ as a similarity metric modulated by a cross-validation gate:

$$S_{align}^{(t)} = \text{Sim}(\mathbf{v}_{init}^{(t)}, \mathbf{v}_{reply}^{(t)}) \cdot \mathcal{V}(M_{init}, M_{reply}), \quad (3)$$

where $\text{Sim}(\cdot)$ is the cosine similarity capturing semantic relevance. The function $\mathcal{V}(\cdot) \in [0, 1]$ serves as a *cross-validation gate*, in which the requestor leverages its own semantic comprehension to assess the factual consistency and logical validity of the response. This multiplicative form ensures that responses with high surface similarity but poor factual validity yield low alignment scores, thereby mitigating the risk of fraud.

Behavioral Consistency: Quantifies the deviation of the current behavior from the historical behavioral profile of agent B . Since LLM agents exhibit intrinsic output stochasticity, a raw deviation measure would conflate benign randomness with malicious strategy shifts. To disentangle these two sources, we introduce a variance-standardized drift measure. Let $\mu_B^{(t-1)}$ be the moving average of agent B 's historical response vectors, and let $\sigma_B^{(t)}$ be the running standard deviation estimate updated via exponential moving average:

$$(\sigma_B^{(t)})^2 = \rho \cdot (\sigma_B^{(t-1)})^2 + (1 - \rho) \cdot \|\mathbf{v}_{reply}^{(t)} - \mu_B^{(t-1)}\|_2^2, \quad (4)$$

where $\rho \in (0, 1)$ is the decay factor. The standardized drift is then defined as:

$$D_{drift}^{(t)} = \max\left(0, \frac{\|\mathbf{v}_{reply}^{(t)} - \mu_B^{(t-1)}\|_2 - \eta \cdot \sigma_B^{(t-1)}}{\sigma_B^{(t-1)}}\right), \quad (5)$$

where $\eta > 0$ is a tolerance coefficient. Only deviations exceeding η standard deviations above the agent's normal behavioral variance produce a positive drift signal. This design accounts for LLM-specific endogenous stochasticity, allowing agents with inherently high output variance to avoid penalization for normal fluctuations, while flagging sudden deviations from a stable behavioral history as potential strategy mutations.

4.1.2. Direct Trust

Direct trust, denoted as T_t^{direct} , represents the subjective assessment of a counterpart's reliability, accumulated from the semantic consistency of historical personal interactions. Since interactions occur sequentially and each round provides partial evidence, we employ Bayesian inference to iteratively update the trust probability as new observations accumulate. Let H be the hypothesis that the target agent is trustworthy, and let $T^{(t)}$ denote the prior probability $P(H)$ at time step t . The evidence E_t for the update consists of two observed semantic signals from the current interaction: semantic alignment $S_{align}^{(t)}$ and standardized drift $D_{drift}^{(t)}$.

Likelihood of Competence: A trustworthy agent is expected to consistently provide high-quality responses. We model the likelihood of observing the current semantic alignment $S_{align}^{(t)}$ as:

$$\begin{cases} P(S_{align}^{(t)} | H) = 1 - e^{-k S_{align}^{(t)}}, \\ P(S_{align}^{(t)} | \neg H) = e^{-k S_{align}^{(t)}}, \end{cases} \quad (6)$$

where $k > 0$ is a sensitivity factor. This exponential form models diminishing returns of trust confidence, ensuring rapid initial learning while saturating at high confidence levels. The ESS conditions derived in Section 4.3 require only that the likelihood function is monotonically increasing and saturating; alternative forms such as logistic or power-law functions satisfy these requirements and yield qualitatively identical equilibria.

Likelihood of Integrity: Stability in semantic behavior is a key indicator of integrity. We model the likelihood using the standardized drift $D_{drift}^{(t)}$:

$$\begin{cases} P(D_{drift}^{(t)} | H) = e^{-\lambda D_{drift}^{(t)}}, \\ P(D_{drift}^{(t)} | \neg H) = 1 - e^{-\lambda D_{drift}^{(t)}}, \end{cases} \quad (7)$$

where $\lambda > 0$ controls the penalty for inconsistency. Trustworthy agents maintain consistent behavior (low standardized drift), while malicious agents exhibit elevated drift due to strategy switching that exceeds their normal behavioral variance.

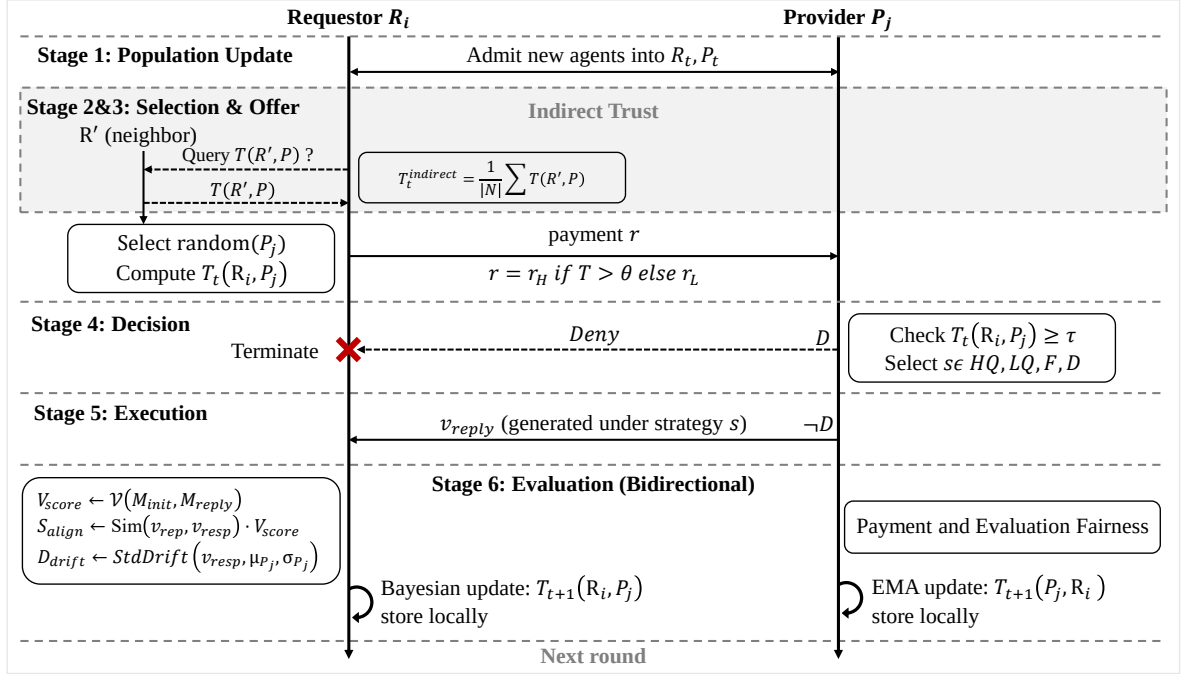


Figure 3: Trust interaction sequence between requestor R_i and provider P_j . Trust updates are asymmetric: $R \rightarrow P$ uses Bayesian posterior update, $P \rightarrow R$ uses EMA. All trust values are stored locally.

Posterior Update: Assuming conditional independence between alignment and drift given the hypothesis, the combined likelihood of evidence E_t is the product of the individual likelihoods. The posterior direct trust for round $t + 1$ is updated using Bayes' theorem:

$$T_{t+1}^{direct} = \frac{P(E_t | H) \cdot T^{(t)}}{P(E_t | H) \cdot T^{(t)} + P(E_t | \neg H) \cdot (1 - T^{(t)})}. \quad (8)$$

In practice, strategy switching may simultaneously increase drift and decrease alignment, introducing positive correlation under $\neg H$. Our standardized drift design mitigates this concern under H : within the normal operating regime (deviations below η standard deviations), $D_{drift}^{(t)}$ is suppressed to zero, substantially reducing spurious correlation with $S_{align}^{(t)}$. Under $\neg H$, the residual correlation causes the independent model to underestimate the joint likelihood $P(E_t | \neg H)$, resulting in a *conservative* approximation error: the trust penalty for malicious agents is weaker than optimal but directionally correct. This conservative property is consistent with the design principle of preferring slower but reliable trust degradation over aggressive but error-prone penalization.

This semantic-aware update mechanism allows the system to rapidly adjust trust based on real-time performance, capturing subtle malicious behaviors such as fraud with high semantic similarity but low factual validity, or behavioral inconsistencies that exceed normal model stochasticity.

4.1.3. Indirect Trust

Indirect trust captures an agent's belief about an unknown partner inferred from the experiences of other agents, and serves primarily as a mechanism to alleviate the cold-start problem in open and dynamic service markets. For a requestor R , the indirect trust in provider P is computed as the average trust held by neighbors who have interacted with P :

$$T_t^{\text{indirect}}(R, P) = \frac{1}{|N_R|} \sum_{R' \in N_R} T_t(R', P), \quad (9)$$

where N_R is the set of requestors in the same interaction neighborhood as R who have previously interacted with P , indirect trust queries occur only between agents that share an interaction neighborhood. Similarly, a provider P evaluates a requestor R using the same aggregation logic to assess reliability before accepting interaction. By combining the Bayesian-updated direct trust with the socially aggregated indirect trust, our measurement framework captures both the intrinsic capability and the social reputation of agents.

The trust update mechanisms are intentionally asymmetric across the two interaction directions. In the $R \rightarrow P$ direction, the requestor observes the full semantic interaction (request and response), enabling the computation of S_{align} and D_{drift} for Bayesian posterior update. In the $P \rightarrow R$ direction, the provider observes only the economic signal (payment amount and evaluation score) and cannot perform semantic cross-validation of its own response. Accordingly, $P \rightarrow R$ trust is updated via exponential moving average: $T_{t+1}(P, R) = \lambda_T \cdot T_t(P, R) + (1 - \lambda_T) \cdot r_{\text{fair}}$, where $r_{\text{fair}} \in [0, 1]$ reflects payment and rating fairness and λ_T is the decay weight. This asymmetry reflects the fundamental difference in information observability between the two directions and does not affect the equilibrium analysis. The complete trust interaction sequence between a requestor and provider is illustrated in Figure 3.

4.2. Expected Revenue

In this stage, we embed trust into the expected revenue function, converting trustworthiness into an economic incentive that drives strategy evolution.

As one of the core elements of game theory, expected revenue is the primary driving force behind the trust mechanism based on the evolution of game strategies in individual agents. Before selecting a strategy, each agent estimates its long-term expected revenue based on the trust level and payment of the counterpart. For a requestor R , the expected revenue aggregates immediate utility and the future value of maintaining access to high-trust providers:

$$U^R = \mathbb{E} \left[\sum_{k=t}^N \gamma^{k-t} u_k^R + \phi^R(T_t^R(P)) \right], \quad (10)$$

where $\gamma \in (0, 1)$ is the discount factor balancing current and future rewards, and N represents the game horizon. The term $\phi^R(T_t^R(P))$ denotes R 's trust influence function, quantifying the capability to identify and select high-quality providers in future rounds. Its specific functional form does not affect the equilibrium analysis, since the stability conditions derived in Section 4.3 depend only on the sign of the requestor's revenue advantage ΔU^R , which holds positive provided the trust mechanism enables rational requestors to identify HQ providers with probability strictly better than random chance.

For a provider P , the expected revenue considers both the service profit and the reputation impact:

$$U^P = \mathbb{E} \left[\sum_{k=t}^N \gamma^{k-t} u_k^P + \phi^P(T_t^R(P)) \right], \quad (11)$$

Here, $\phi^P(T)$ represents the future reward potential brought about by the current trust value. A higher trust increases the probability of being selected for future transactions, translating reputation into tangible economic returns. We model this influence as:

$$\phi^P(T) = \alpha \cdot \psi(T) \cdot \bar{r}_T, \quad (12)$$

where $\alpha > 0$ is a weighting parameter representing the importance of future engagement. $\psi(T)$ is the market selection probability function (e.g., a Sigmoid function), monotonically increasing in T ($\psi'(T) > 0$), indicating that agents with higher trust have a higher probability of being selected. The term $\bar{r}_T$ denotes the expected payment level under the threshold policy: $\bar{r}_T = r_H$ for agents with $T > \theta$, and $\bar{r}_T = r_L$ otherwise.

Revenue Advantage and Mean-Field Approximation. To connect the individual expected revenue to the population-level evolutionary dynamics analyzed in Section 4.3, we derive the expected revenue advantage of cooperative over malicious strategies under a mean-field approximation. Specifically, let $u_{HQ} = \mathbb{E}_i[U_i^P \mid s_i = HQ]$ and $u_F = \mathbb{E}_i[U_i^P \mid s_i = F]$ denote the population-average expected revenues for providers adopting HQ and F strategies, respectively. This approximation treats individual variation in interaction history as averaged out at the population level, and is valid when the population is sufficiently large and interactions are sufficiently mixed.

Let T_{high} denote the steady-state average trust of providers who consistently adopt HQ, and T_{low} denote that of providers who adopt F. Substituting the revenue components from Eq. (11), the provider's revenue advantage is:

$$\begin{aligned} \Delta U^P &= u_{HQ} - u_F \\ &= \underbrace{(\delta - c_h)}_{\text{Immediate margin diff.}} + \underbrace{\alpha [\psi(T_{high}) \cdot r_H - \psi(T_{low}) \cdot r_L]}_{\text{Trust-induced future gain}}. \end{aligned} \quad (13)$$

Under the conservative assumption $\bar{r}_{T_{high}} \approx \bar{r}_{T_{low}} \approx r_H$ (even low-trust agents may occasionally receive high payments), this simplifies to:

$$\Delta U^P = (\delta - c_h) + \alpha \cdot r_H \cdot \Delta \Psi_{\text{eff}}, \quad (14)$$

where $\Delta \Psi_{\text{eff}} = \psi(T_{high}) - \psi(T_{low})$ represents the effective difference in market selection probability between high-trust and low-trust agents. The first term $(\delta - c_h) < 0$ captures the immediate cost disadvantage of honest service (since $c_h > \delta$ as established in Section 3.1). The second term represents the long-term trust-induced gain that can overcome this disadvantage. This revenue structure serves as the fitness function for the population-level evolutionary dynamics analyzed in the following section.

4.3. Game Equilibrium Theoretical Analysis

In this stage, we introduce the **Evolutionarily Stable Strategy (ESS)** to define stability in decentralized multi-agent scenarios: a state where dominant cooperative strategies cannot be invaded by a fraction of mutant strategies (e.g., fraud). To formally characterize this stability, we employ the Replicator Dynamics, which capture the population-level evolution of strategies, and derive the boundary conditions for maintaining a robust cooperative equilibrium.

Micro-Foundation of Replicator Dynamics: The Replicator Dynamics describe macro-level strategy evolution but require a micro-level behavioral justification for LLM-based agents. In our framework, each agent selects a strategy based on its cumulative expected revenue following a noisy best response rule (McKelvey and Palfrey, 1995): the probability of selecting strategy s is proportional to $\exp(\beta \cdot U_s)$, where $\beta > 0$ governs the degree of rationality. This logit choice model naturally captures the behavior of LLM agents, whose strategy decisions are guided by expected revenue information but subject to intrinsic generation stochasticity, with the LLM temperature parameter playing an analogous role to $1/\beta$. When a large population of agents independently updates strategies via noisy best response, the resulting population-level dynamics converge to the Replicator Dynamics (Sandholm, 2010), providing the formal micro-foundation for our analysis without requiring biological evolutionary assumptions.

Stability Analysis: Let $x \in [0, 1]$ and $y \in [0, 1]$ denote the population fractions of Normal Providers ($\mathcal{P}_n$) and Normal Requestors ($\mathcal{R}_n$), respectively. The system evolution is governed by the Replicator Dynamics:

$$\dot{x} = x(1-x)\Delta U^P, \quad \dot{y} = y(1-y)\Delta U^R, \quad (15)$$

where ΔU^P and ΔU^R represent the expected revenue advantages of normal strategies over malicious ones.

To address realistic constraints such as observation latency and incomplete information inherent in decentralized settings, we model the revenue difference conservatively. Substituting the revenue composition from Eq. (10) and Eq.(11), the provider's revenue advantage is derived as:

$$\Delta U^P = \underbrace{(\delta - c_h)}_{\text{Immediate Advantage Diff}} + \underbrace{\alpha r_H \cdot \Delta \Psi_{\text{eff}}}_{\text{Effective Trust Gain}}, \quad (16)$$

Here, $\Delta \Psi_{\text{eff}}$ represents the *effective expected difference* in market selection probability between high-trust and low-trust agents. This term implicitly accounts for observation noise: if the trust mechanism fails to distinguish behavior due to significant delays or noise, $\Delta \Psi_{\text{eff}}$ approaches zero. For requestors, $\Delta U^R > 0$ typically holds provided that the mechanism enables rational agents to identify HQ providers with a probability strictly better than random chance.

With ΔU^P and ΔU^R specified in Eqs. (15)–(16), we next study the local stability of the fully cooperative equilibrium $(x^*, y^*) = (1, 1)$ under the Replicator Dynamics. To determine stability, we analyze the Jacobian matrix J of the system at the cooperative equilibrium $(x^*, y^*) = (1, 1)$. Since the cross-derivatives vanish at pure strategy endpoints, the eigenvalues of J are strictly determined by the negative revenue differences:

$$\lambda_1 = \left. \frac{\partial \dot{x}}{\partial x} \right|_{(1,1)} = -\Delta U^P, \quad \lambda_2 = \left. \frac{\partial \dot{y}}{\partial y} \right|_{(1,1)} = -\Delta U^R, \quad (17)$$

According to Lyapunov Stability Theory (Khalil and Grizzle, 2002), the state $(1, 1)$ is asymptotically stable if and only if all eigenvalues are negative, which requires $\Delta U^P > 0$ and $\Delta U^R > 0$. Substituting Eq. (16) into $\Delta U^P > 0$ yields an explicit threshold on the future weight α , which establishes the following theorem.

Theorem 4.1 (*Ev-Trust Evolutionary Stability Condition*). *Consider the Replicator Dynamics in Eq. (15) with payoff advantages ΔU^P and ΔU^R , where $\Delta U^P = (\delta - c_h) + \alpha r_H \Delta \Psi_{\text{eff}}$ as in Eq. (16). If $\Delta U^R > 0$ and*

$$\alpha > \frac{c_h - \delta}{r_H \Delta \Psi_{\text{eff}}}, \quad (18)$$

Then the fully cooperative equilibrium $(x^, y^*) = (1, 1)$ is globally stable and constitutes an Evolutionarily Stable Strategy (ESS), hence robust to invasion by a small fraction of mutant strategies.*

Remark 4.2. Notably, Eq. (18) indicates that $\Delta \Psi_{\text{eff}}$ controls the robustness margin: when observation noise or delays weaken trust differentiation, $\Delta \Psi_{\text{eff}}$ decreases, and the stability threshold on α increases accordingly.

In general, cooperation remains dominant when the long-term reputation gain $\alpha r_H \Delta \Psi_{\text{eff}}$ exceeds the short-term defection incentive $(c_h - \delta)$. Furthermore, the cooperative equilibrium is not merely locally stable but globally attracting: any initial state with $x_0 > 0$ and $y_0 > 0$ asymptotically converges to $(1, 1)$ under the conditions of Theorem 4.1 (see Appendix C for details of the global stability proof).

5. EXPERIMENTS

5.1. Experimental Settings

Scenario Configuration. We implement a decentralized multi-agent service market to evaluate *Ev-Trust*. The system consists of N agents driven by heterogeneous LLMs (*gpt-5-nano*, *gpt-4.1-nano*, *gpt-4o-mini*), the types of which vary as shown in Table 1. This heterogeneous configuration reflects the realistic diversity of model capabilities across autonomous agents. Each agent’s LLM is responsible for analyzing expected revenue, selecting interaction strategies, generating task requests and responses, and evaluating service quality. Numerical computations, including trust updates and revenue calculations, are handled by deterministic program logic. Service tasks are drawn from the TruthfulQA dataset (Lin et al., 2022), which provides 817 factual questions with annotated correct and common incorrect answers, supplemented by TriviaQA (Joshi et al., 2017) for additional scale. TruthfulQA is particularly suited to our setting as it enables controlled evaluation of masquerade providers (P_{m1}), who can exploit the annotated incorrect answers as fraudulent responses. Provider selection follows softmax matching based on composite trust scores, with selection intensity β_s controlling the degree of trust-based differentiation. The complete experimental parameter configuration, along with theoretical justification of parameter selection, is provided in Appendix A. Specific prompts for each agent role are provided in Appendix F.

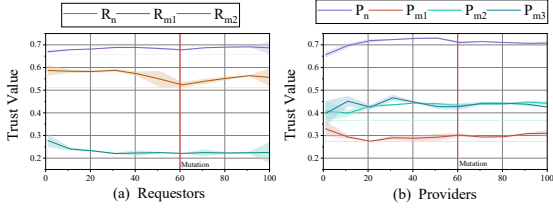


Fig. 4. Trust Evolution of Different Agent Types.

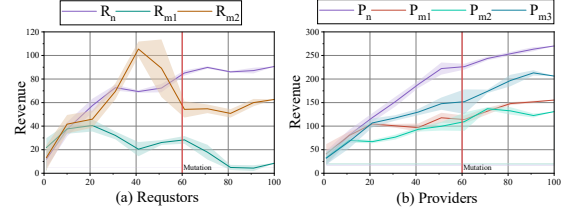


Fig. 5. Revenue Trend of Different Agent Types.

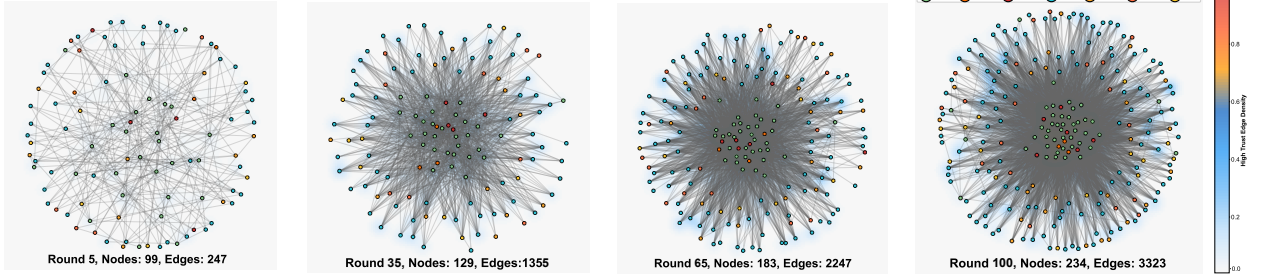


Figure 6: Network Topology Evolution. Higher trust shortens node distance. Trust values are inversely proportional to edge lengths; warm-colored regions indicate clusters of high trust.

Population Dynamics. The market simulates a dynamic open environment: at each round, new agents equivalent to 2% of the current population enter the system with random types. The simulation spans $M = 100$ rounds with initial trust $T_0 = 0.5$, and controlled Gaussian noise (up to 5%) is introduced into transaction payments to simulate real-world volatility. In round 60, a mutation event injects additional malicious participants equivalent to 20% of the population, testing resilience against sudden adversarial perturbation. All experiments are repeated over 5 independent random seeds, with results reported as mean $\pm$ standard deviation. For evaluation purposes, the reported trust level of an agent is computed externally as the mean of all local trust values held by its interaction partners, without being accessible to any agent.

Baselines. We compare *Ev-Trust* against three baselines spanning distinct paradigms for regulating decentralized agent interactions. All baselines are integrated into an identical interaction loop, aligning their input signals and output decisions with our system’s metrics to ensure a consistent evaluation ground. (1) **EigenTrust** (Kamvar et al., 2003): A classic trust aggregation method that computes global trust scores via transitive propagation. We adapt it by normalizing local service ratings into the transition matrix. Unlike *Ev-Trust*’s purely local trust computation, EigenTrust requires network-wide iterative convergence. (2) **LR2** (Ren et al., 2025): A recent decentralized reputation mechanism based on multi-agent reinforcement learning. Each agent maintains two REINFORCE policies: a dilemma policy for strategy selection and an evaluation policy for assigning reputation scores to neighbors. Reputation scores reshape payoffs via reward shaping. We use a 20-round burn-in period for policy warm-up. (3) **Fermi Update Rule** (Zhang et al., 2024a): A classical evolutionary baseline where strategy updates are driven purely by revenue-based imitation rather than trust-based selection. Each agent randomly selects a neighbor and adopts its strategy with probability inversely related to their revenue difference, governed by selection intensity κ . Trust values are computed and recorded as observational metrics, but do not influence partner selection or strategy decisions. This baseline isolates the effect of removing trust-driven screening from the evolutionary process.

5.2. Experimental results

In this stage, we validate the effectiveness of *Ev-Trust* from both empirical and theoretical perspectives. First, we present empirical evidence regarding trust, utility, and network topology; then, through phase-space analysis, we demonstrate the correspondence between these empirical phenomena and the predictions of Theorem 4.1.

Trust Evolution of Different Agents. As shown in Figure 4, the trust values of normal agents (R_n , P_n) and malicious agents are correctly separated from the early rounds of the simulation, with normal agents consistently maintaining higher trust than all malicious types despite minor fluctuations. Among the malicious agents, R_{m2}

exhibits trust values close to R_n due to its non-adversarial but noisy evaluation behavior. On the provider side, the strategy-switching agent P_{m3} shows initial volatility from its transition to exploitative strategies; however, the variance-standardized drift measure D_{drift} rapidly detects this behavioral shift, driving its trust persistently low. The masquerade agent P_{m1} is similarly suppressed despite its high semantic similarity, confirming the effectiveness of the cross-validation gate $\mathcal{V}(\cdot)$ in detecting factually flawed responses.

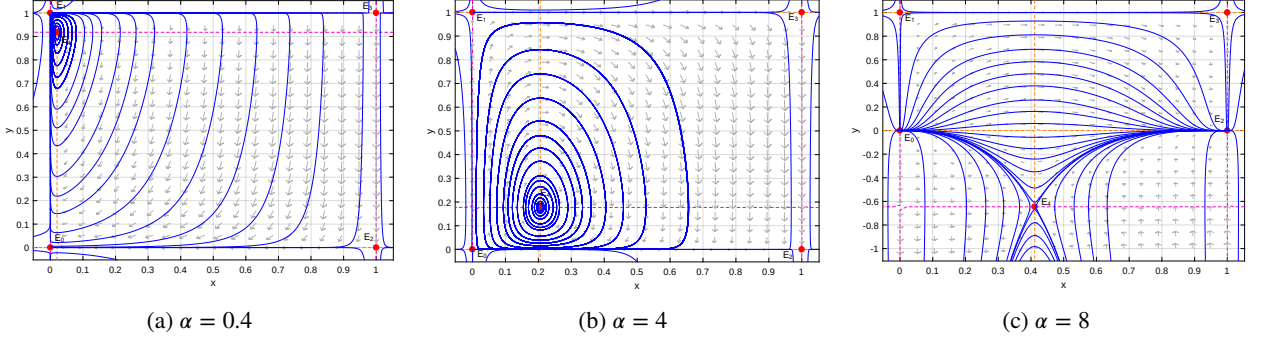


Figure 7: Phase portraits of *Ev-Trust* generated by MATLAB pplane9 (Dr. John C. Polking). The vector field indicates strategy trajectories under different values of α .

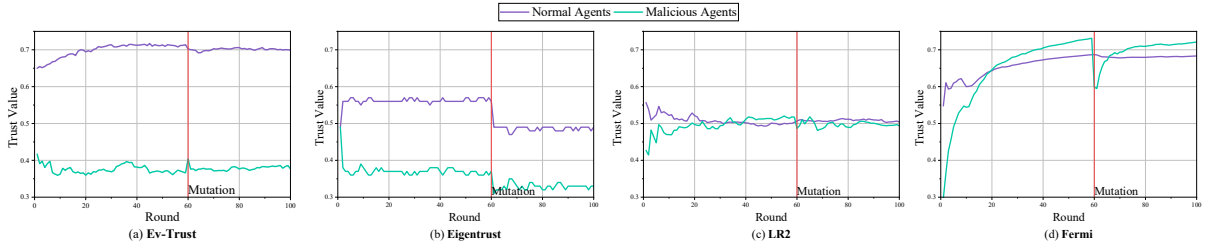


Figure 8: Trust evolution under malicious mutation. The red line marks the mutation injection at round 60.

Revenue Trend of Different Agents. As shown in Figure 5, normal requestors and providers exhibit steadily increasing revenue that approaches an equilibrium level, reflecting the compounding benefit of sustained high trust and preferential selection. All malicious agent types are economically suppressed, with revenues consistently below those of normal agents. Notably, R_{m2} briefly surpasses R_n in early rounds due to randomness in partner selection, but its revenue growth stagnates and falls below R_n as the trust mechanism matures. These results confirm that the trust-evolution coupling effectively converts trustworthiness into sustained economic advantage.

Network Topology Cluster Evolution. Figure 6 visualizes the trust network at four stages (rounds 5, 35, 65, and 100), where spatial proximity reflects trust levels. The network initially exhibits a loosely connected, undifferentiated structure. As the system evolves, normal agents progressively aggregate into cohesive high-trust clusters (warm-colored regions), while malicious agents are pushed to the sparse periphery. By round 100, a clear core-periphery structure emerges: the cooperative core is tightly connected, while malicious agents are effectively isolated. This structural evolution demonstrates that the trust mechanism reshapes the network topology to favor cooperative clusters.

Theoretical Convergence Analysis. To verify the threshold effect predicted by Theorem 4.1, we analyze the phase portraits of the Replicator Dynamics in Eq. (15) under three representative values of α : below the theoretical threshold ($\alpha = 0.4 < 0.88$), at the main experimental setting ($\alpha = 4$), and above ($\alpha = 8$). Figure 7 visualizes the transition using a normalized effective incentive coefficient. When $\alpha = 0.4$, the interior equilibrium at $(0.0206, 0.9177)$ indicates a weak-incentive regime where cooperation depends on restrictive initial conditions. When $\alpha = 4$, the interior threshold moves to $(0.2057, 0.1771)$, enlarging the cooperative attraction region. When $\alpha = 8$, the interior intersection moves outside the feasible strategy space at $(0.4114, -0.6458)$, so the vector field is dominated by convergence toward $(1, 1)$. These results show that increasing trust-revenue incentives shifts or removes the internal barrier to cooperation, consistent with Theorem 4.1 and explaining why *Ev-Trust* promotes the dominance of normal agents in the main experiments.

5.3. Method Comparison

In this stage, we adapt various baseline methods to the context of interactions with decentralized LLM services and evaluate them based on three criteria: trust discrimination, strategy suppression, and average utility.

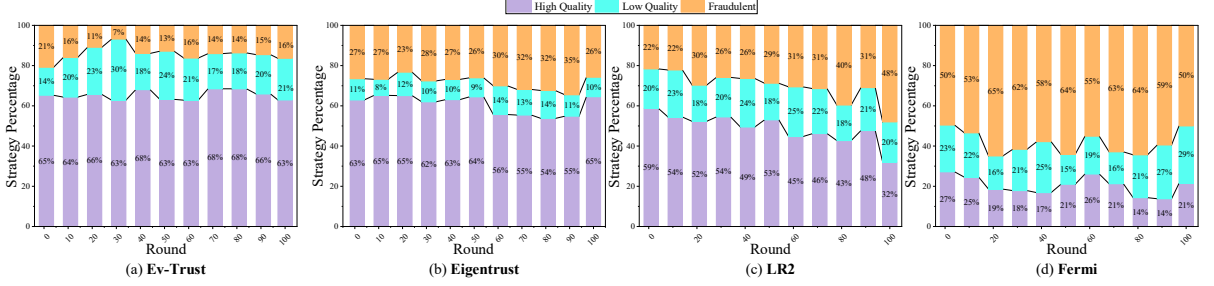


Figure 9: Strategy Evolution. The proportions of the three strategies in all interactions change as the system evolves. Service denials are excluded because they do not directly affect trustworthiness evaluation.

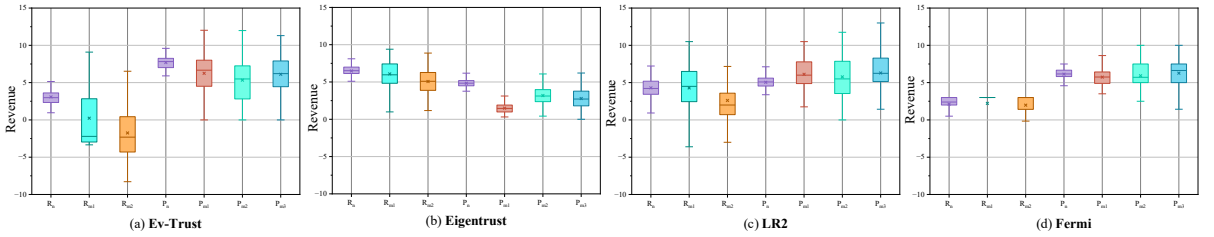


Figure 10: Agent revenue comparison. The figure compares the revenue of different agent types under *Ev-Trust* and baseline mechanisms.

Trust Value Evolution. All four methods are compared by plotting the average trust of normal versus malicious agents over 100 rounds (Figure 8). *Ev-Trust* achieves the highest trust separation, with the gap widening toward a clear equilibrium. Following the mutation injection, *Ev-Trust* exhibits a brief dip but rapidly recovers, demonstrating strong resilience. EigenTrust achieves reasonable separation during normal operation, but the mutation event causes a severe and sustained trust collapse. LR2 and Fermi both exhibit limited trust separation throughout, though both show moderate resilience to the mutation event.

Strategy Evolution. To assess the impact of each method on strategy decisions, we tracked the proportions of HQ, LQ, and F strategies every 10 rounds (Figure 9). Under *Ev-Trust*, the HQ proportion is the highest and remains stable, while fraudulent behaviors are significantly suppressed. EigenTrust fails to curb fraudulent strategies, with F maintaining a disproportionately high share. LR2 and Fermi suffer from a gradual decline in HQ proportion as malicious agents capture more interaction opportunities. The Fermi baseline exhibits the most severe degradation: normal providers imitate the temporarily high revenues of fraudulent agents, causing the HQ proportion to deteriorate rapidly.

Game Revenue. Figure 10 compares the per-agent-type average revenue across all 100 rounds using box plots. Under *Ev-Trust*, normal agents consistently surpass all malicious types in both mean revenue and stability. Under EigenTrust, malicious requestors achieve higher revenue than normal requestors, indicating a failure of the incentive structure on the requestor side. Both LR2 and Fermi exhibit scenarios where malicious agents economically outperform normal agents, reversing the intended incentive hierarchy. Only *Ev-Trust* successfully ensures that trustworthiness translates into a decisive economic advantage across both roles.

Malicious Agent Participation Rate. As detailed in Table 3, *Ev-Trust* achieves the lowest malicious participation rates across nearly all agent types, indicating that the trust-based softmax selection mechanism effectively excludes low-trust agents from the interaction pool. In contrast, EigenTrust, LR2, and Fermi all permit substantially higher malicious participation, with malicious providers accounting for over 20% of interactions under LR2 and Fermi.

Table 3

Percentage of interactions involving malicious agents. The selection rate represents the proportion of a specific type of malicious agent participating in all interactions.

Method	Requestors		Providers		
	R_{m1}	R_{m2}	P_{m1}	P_{m2}	P_{m3}
Eigentrust	11.76%	15.59%	22.34%	12.13%	10.84%
LR2	12.12%	10.24%	21.99%	11.71%	12.21%
Fermi	11.76%	15.40%	22.58%	11.40%	11.62%
<i>Ev-Trust</i>	<u>6.69%</u>	<u>9.25%</u>	<u>10.18%</u>	<u>5.09%</u>	<u>7.69%</u>

Table 4

Game Parameter Sensitivity. Default: $\alpha_{threshold}=0.88$ $c_h=5$, $\delta=1$, $r_H=15$.

Regime	$\bar{T}(P_n)$	$\bar{T}(P_{m1})$	Trust Diff.	HQ Ratio	F Ratio	$\bar{U}(P_n)$	Mal. Select.
Default ($\alpha=4.0$)	0.725	0.292	0.434	68%	10.6%	194.4	36.3%
Low α ($\alpha=0.4$)	0.52	0.275	0.244	13.2%	12.9%	148.7	39.2%
High α ($\alpha=8.0$)	0.759	0.288	0.47	56.6%	12%	179.2	34.7%
High cost ($c_h=7$)	0.686	0.291	0.395	64.7%	10.5%	150.1	39.6%
Low penalty ($\delta=0.5$)	0.762	0.276	0.485	66.7%	11.2%	171.2	40.6%
Low payment ($r_H=12$)	0.685	0.284	0.4	63.2%	13.6%	151.4	39.2%

5.4. Sensitivity Analysis

5.4.1. Game Parameter Sensitivity

To verify that the cooperative equilibrium is robust to economic variations, we evaluate *Ev-Trust* under five parameter regimes, each modifying a single parameter while keeping others at default. **Low α** sets the future engagement weight below the theoretical stability threshold (Eq. 18), while **High α** doubles the default to test convergence acceleration. The remaining three regimes intensify the incentive to defect: **High cost** ($c_h = 7$) compresses honest profit margin, **Low penalty** ($\delta = 0.5$) makes fraud more immediately profitable, and **Low payment** ($r_H = 12$) weakens the trust-induced future advantage. As shown in Table 4, the Low α regime fails to achieve effective trust differentiation, empirically confirming the theoretical threshold. The High α regime achieves incrementally stronger separation, indicating that the default $\alpha = 4.0$ already operates well within the Cooperative Regime. Among the three economic regimes, **Low penalty** exhibits the largest degradation in trust separation, as the reduced fraud cost directly narrows the gap $c_h - \delta$ that the trust-induced future gain must overcome. Under all regimes except Low α , the mechanism maintains positive trust differentiation and HQ-dominant strategy composition.

5.4.2. Trust Parameter Sensitivity

To assess robustness to hyperparameter selection, we vary each of the five key trust parameters individually, testing one value above and one below the default. As shown in Table 5, the direct trust weight ω and selection intensity β_s have the strongest influence on system behavior: reducing ω amplifies the influence of malicious recommenders, and reducing β_s to 1.0 makes selection nearly random, diminishing the screening power of trust. The alignment sensitivity k and drift penalty λ primarily affect convergence speed. The drift tolerance η controls the trade-off between detecting genuine strategy shifts and tolerating normal LLM variance. Across all configurations, the system maintains positive trust differentiation and HQ-dominant strategy proportions, confirming that the mechanism is not critically dependent on precise hyperparameter tuning.

5.5. Ablation Study

To validate the necessity of the two key LLM-specific designs in *Ev-Trust*, we conduct two ablation experiments, each disabling one component while keeping the rest intact.

No-Gate Ablation. The cross-validation gate $\mathcal{V}(\cdot)$ is disabled by fixing $\mathcal{V} = 1.0$, so that semantic alignment reduces to raw cosine similarity. As shown in Table 6, removing the gate causes the trust of P_{m1} to rise substantially, approaching that of P_n , and trust differentiation collapses. This confirms that cosine similarity alone cannot distinguish factually flawed but semantically relevant responses from genuinely accurate ones, validating the necessity of the cross-validation gate for countering vulnerability (i).

Table 5
Trust Parameter Sensitivity.

Param.	Value	$\bar{T}(P_n)$	$\bar{T}(P_{m1})$	Trust Diff.	HQ Ratio	F Ratio	$\bar{U}(P_n)$	Mal. Select.
Default	Table A.1	0.725	0.292	0.434	68%	10.6%	194.4	36.3%
ω	0.3	0.757	0.28	0.476	66.3%	10.3%	171	35.9%
	0.9	0.752	0.294	0.458	63.4%	10.3%	168.2	35%
k	0.3	0.66	0.297	0.363	67.7%	13.4%	159.9	29.9%
	1.5	0.796	0.293	0.502	68.6%	11%	162.4	38.4%
λ	0.5	0.794	0.285	0.509	64.5%	10.9%	157.2	32.3%
	3.0	0.67	0.292	0.378	65.4%	12.9%	151.6	36.2%
η	0.5	0.538	0.275	0.263	66.7%	13.3%	148.2	42.1%
	2.5	0.788	0.294	0.494	70%	6.67%	168.6	31%
β_s	1.0	0.719	0.274	0.445	63.3%	6.67%	137.8	36.9%
	10.0	0.743	0.281	0.461	68%	13.2%	180.5	37.1%

Table 6
Ablation Study. Default denotes the full *Ev-Trust* mechanism.

Configuration	$\bar{T}(P_n)$	$\bar{T}(P_{m1})$	Trust Diff.	HQ Ratio	F Ratio	$\bar{U}(P_n)$	Mal. Select.
Default (full <i>Ev-Trust</i>)	0.725	0.292	0.434	68%	10.6%	194.4	36.3%
No-Gate ($\mathcal{V} \equiv 1.0$)	0.748	0.472	0.276	64.5%	19.6%	161.3	34.5%
No-Evolution ($\alpha = 0$)	0.487	0.287	0.2	32.6%	22.5%	144.3	43.5%

No-Evolution Ablation. The trust-evolution coupling is disabled by setting $\alpha = 0$, so that trust values are computed but do not feed into the expected revenue function. As shown in Table 6, malicious agents maintain high participation rates, and the HQ ratio fails to increase despite trust values correctly differentiating agent types. This demonstrates that accurate trust assessment is necessary but not sufficient: without embedding trust into the revenue structure, agents lack the economic incentive to cooperate, and the population does not converge to the equilibrium predicted by Theorem 4.1.

6. CONCLUSION

This paper proposed *Ev-Trust*, a trust mechanism grounded in evolutionary game theory for decentralized LLM-based multi-agent service economies. To address three vulnerabilities of LLM-based services, namely near-zero fraud cost, difficulty in semantic quality verification, and endogenous output stochasticity, *Ev-Trust* introduces three targeted designs: a cross-validation gate for semantic quality assessment, a variance-standardized drift measure for behavioral anomaly detection, and a trust-revenue coupling that embeds trust signals into the expected revenue function. Theoretically, we proved the local stability and global attraction of the cooperative equilibrium based on Replicator Dynamics with a noisy best response micro-foundation, and derived an explicit threshold condition requiring that long-term trust-induced gain exceeds the short-term fraud incentive. Extensive experiments with heterogeneous LLM agents confirmed that *Ev-Trust* effectively suppresses malicious strategies, maintains correct economic incentive hierarchies,

and exhibits resilience against adversarial perturbation, consistently outperforming all baselines in trust differentiation, strategy composition, and revenue distribution.

6.1. Theoretical Implications

This study contributes to the literature on trust mechanisms and multi-agent evolutionary dynamics in two respects.

First, it bridges semantic trust evaluation with population-level evolutionary stability analysis. Traditional trust mechanisms (Kamvar et al., 2003; Josang and Ismail, 2002) treat trust as a passive reputation aggregator decoupled from strategy dynamics, while existing evolutionary studies (Han, 2022; Sarkadi and Lewis, 2024) model strategy evolution without grounding the fitness function in observable interaction signals. By embedding LLM-specific trust signals into the expected revenue function and analyzing the resulting Replicator Dynamics, *Ev-Trust* establishes a unified framework in which trust functions not as a passive record but as an active evolutionary force.

Second, this work provides a formal micro-foundation linking the stochastic decision behavior of LLM agents to macro-level evolutionary dynamics through noisy best response (McKelvey and Palfrey, 1995; Sandholm, 2010), justifying the application of evolutionary stability analysis to LLM agent populations without invoking biological evolutionary assumptions.

6.2. Practical Implications

The *Ev-Trust* framework offers practical value for deploying trustworthy LLM-based agents in decentralized service environments. By relying solely on locally observable semantic signals and locally maintained trust states, the framework eliminates the dependence on centralized supervision or privileged access to agent internals that limit existing LLM trust mechanisms (Wang et al., 2026; Gao et al., 2025), enabling deployment across heterogeneous platforms without a global reputation authority.

The trust-revenue coupling further provides a self-regulating economic mechanism that suppresses malicious strategies at the population level without requiring external enforcement. As autonomous systems evolve toward agentic workflows, cross-platform service orchestration, and decentralized AI economies, this trust-evolution coupling mechanism offers an extensible foundation for sustaining cooperation in these emerging settings.

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Appendix

A. Justification and Explanation of Parameters

This appendix provides the complete experimental parameter configuration along with theoretical justification for each parameter group. Table A.1 summarizes all parameter values used in the experiments, organized into four categories: population and simulation settings, decision thresholds, trust hyperparameters, and game payoff parameters. All values are selected to satisfy fundamental economic constraints and game-theoretic stability conditions detailed in the following subsections.

Table A.1
Experimental Parameter Configuration.

Param.	Description	Value	Source
N	Initial agent population ($N_R=35, N_P=65$)	100	Config
π_R, π_P	Type ratios for R and P	See note	Config
M, T_0	Simulation rounds; initial trust	100; 0.5	Config
ω	Direct trust weight	0.7	Eq. (2)
θ, τ	Payment / interaction thresholds	0.4; 0.4	Workflow 3.2
β_s	Selection intensity	5.0	Workflow 3.2
α, γ	Future weight; discount factor	4.0; 0.85	Eq. (18), (10)
k, λ	Alignment / drift sensitivity	0.65; 1.35	Eq. (6), (7)
η, ρ	Drift tolerance; variance decay	1.4; 0.8	Eq. (5), (4)
r_H, r_L	High / low payment	15; 10	Table 2
c_h, c_l	HQ / LQ execution cost	5; 4	Table 2
δ	Fraud risk cost	1	Table 2
u_h, u_l	HQ / LQ service utility	20; 11	Table 2
κ	Fermi selection intensity	0.6	Baseline config

Note: $\pi_R = (0.8, 0.1, 0.1)$ for (R_n, R_{m1}, R_{m2}) ; $\pi_P = (0.6, 0.2, 0.1, 0.1)$ for $(P_n, P_{m1}, P_{m2}, P_{m3})$.

A.1. Game Matrix Constraints

For the service market to function effectively, the system must adhere to Rational Participation Constraints, including Individual Rationality (IR) and Incentive Compatibility (IC). The three constraints specified in Section 3.2 are restated here with their economic interpretation:

(1) Profitability Constraint: $r_H > c_h > c_l > 0$. This ensures that honest providers maintain a positive margin under high payment. With our experimental values $r_H = 15$, $c_h = 5$, $c_l = 4$, all honest strategies yield strictly positive immediate revenue.

(2) Participation Constraint: $u_h > r_H$ and $u_l > r_L$. This ensures that requestors benefit from participating in the market. With $u_h = 20 > r_H = 15$ and $u_l = 11 > r_L = 10$, requestors have non-negative utility from both HQ and LQ services.

(3) Fraud Incentive Constraint: $c_h > \delta$, i.e., $5 > 1$. This means the provider's immediate margin from fraud ($r_H - \delta = 14$) exceeds that from honest service ($r_H - c_h = 10$). This condition is the source of the short-term fraud incentive that the trust mechanism must counteract through the long-term trust-induced gain $\alpha r_H \Delta \Psi_{\text{eff}}$ (Theorem 4.1).

A.2. Decision Thresholds

Two thresholds govern the decision process, each optimizing a different aspect of the interaction.

Payment threshold $\theta = 0.4$ defines the boundary separating high-payment from low-payment offers, balancing two competing risks: a higher θ reduces false positives (overpaying untrustworthy providers) but increases false negatives (underpaying trustworthy ones). With initial trust $T_0 = 0.5 > \theta$, new agents receive high payment by default, facilitating cold-start participation.

Interaction threshold $\tau = 0.4$ acts as a risk-control filter for providers. A provider accepts a request only if its trust in the requester exceeds τ , thereby reducing exposure to unreliable requesters.

$$(1 - \tau) \cdot (-r_H) + \tau \cdot (u_h - r_H) > 0. \quad (\text{A.1})$$

Substituting our values: $(1 - 0.4) \cdot (-15) + 0.4 \cdot (20 - 15) = -9 + 2 = -7$, which is negative under the worst-case interpretation. In practice, τ operates as a minimum quality filter: providers reject requestors whose historical behavior suggests unreliable payment, while the positive expected revenue from the majority of interactions compensates for occasional losses.

A.3. Trust Hyperparameters

Six parameters control the responsiveness and robustness of the trust evaluation system.

Alignment sensitivity $k = 0.65$ and drift penalty coefficient $\lambda = 1.35$ govern the steepness of the Bayesian likelihood functions (Eqs. (6)–(7) in the main text). As established in Section 4.1, ESS conditions require only monotonicity and saturation. Specific values are calibrated to balance responsiveness (rapid trust differentiation in early rounds) with stability (resistance to noise-induced false updates). Sensitivity analysis in Section 5.4 confirms robustness across $k \in [0.3, 1.5]$ and $\lambda \in [0.5, 3.0]$.

Drift tolerance $\eta = 1.4$ determines how many standard deviations of normal behavioral variance an agent can exhibit before triggering a positive drift signal. This value follows anomaly detection conventions where thresholds between 1 and 2 standard deviations balance sensitivity against false positive rates.

Variance decay factor $\rho = 0.8$ controls the memory of the running variance estimate, with higher values giving more weight to historical variance. A value of 0.8 provides a half-life of approximately 3 rounds, aligning with typical interaction frequencies.

Direct trust weight $\omega = 0.7$ ensures that personal interaction experience dominates over neighbor recommendations, limiting the impact of malicious recommenders (R_{m1}) to the cold-start phase. Sensitivity analysis confirms that values in $[0.5, 0.9]$ all maintain effective trust differentiation.

Selection intensity $\beta_s = 5.0$ governs softmax provider selection in the workflow. With a trust difference of 0.3 between two providers, this value causes the higher-trust provider to be selected approximately 4.5 times more often. Sensitivity analysis confirms that values in $[3.0, 10.0]$ maintain effective screening, while $\beta_s = 1.0$ approaches random selection and degrades performance.

A.4. Cooperation Boundary

Cooperative equilibrium stability depends on the relationship between immediate fraud incentives and long-term trust-induced gains, formalized by the Cooperation Boundary:

$$B(\alpha, \delta) = \alpha \cdot r_H \cdot \Delta\Psi_{\text{eff}} - (c_h - \delta). \quad (\text{A.2})$$

Two regimes emerge from this boundary. In the Cooperative Regime ($B > 0$), long-term trust gain exceeds short-term fraud incentive, driving the system toward cooperation. In the Collapse Regime ($B < 0$), fraud incentive dominates, leading to market failure. From Theorem 4.1, the critical threshold is $\alpha > (c_h - \delta)/(r_H \cdot \Delta\Psi_{\text{eff}})$. With experimental values $c_h = 5$, $\delta = 1$, $r_H = 15$, and $\Delta\Psi_{\text{eff}} \approx 0.3$ (estimated from converged trust values), this yields $\alpha > 4/(15 \times 0.3) \approx 0.89$. Our setting $\alpha = 4.0$ places the system firmly in the Cooperative Regime, as empirically validated by the α threshold verification experiment in Section 5.4.

B. The Analysis of Computational Complexity

A critical requirement for decentralized systems is scalability. We analyze the complexity for a single agent i interacting with the market in one round t , where $|\mathcal{A}|$ denotes the total population size, $|\mathcal{P}|$ denotes the number of providers, and $|N_i|$ represents the size of agent i 's interaction neighborhood (the set of agents with whom i has previously interacted).

Space Complexity. Each agent maintains a local trust vector over its interaction partners and a limited history of response embeddings for drift computation, yielding a storage requirement of $\mathcal{O}(|N_i|)$. Since each agent interacts with only a subset of the population over time, $|N_i| \ll |\mathcal{A}|$. The variance-standardized drift tracker (Eqs. (4)–(5)) requires two additional scalars per partner (μ_B and σ_B^2), which do not change the asymptotic bound. Consequently, the space requirement scales linearly with the number of interaction partners rather than the square of the total population.

Time Complexity. The per-round workflow consists of six phases with the following costs for a single requestor i :

(1) *Selection.* Computing softmax probabilities over all candidate providers requires evaluating $T_i(i, j)$ for each $j \in \mathcal{P}$, yielding $\mathcal{O}(|\mathcal{P}|)$. For providers with prior interaction history, the composite trust $T_i(i, j) = \omega \cdot T_i^{\text{direct}}(i, j) + (1 - \omega) \cdot T_i^{\text{indirect}}(i, j)$ is retrieved in $\mathcal{O}(1)$; for cold-start providers, indirect trust aggregation requires $\mathcal{O}(|N_i|)$.

(2) *Offer and Decision.* Threshold comparisons against θ and τ , each $\mathcal{O}(1)$.

(3) *Execution.* Response generation and cross-validation via $\mathcal{V}(\cdot)$ involve LLM API calls. These are constant-time from the mechanism's perspective ($\mathcal{O}(1)$ per interaction), though wall-clock latency depends on model inference speed. Embedding computation for S_{align} is also $\mathcal{O}(1)$ per interaction when using a local embedding model with pre-cached question embeddings.

(4) *Evaluation.* The Bayesian posterior update (Eq. (8)) involves computing two likelihood products and a division, all $\mathcal{O}(1)$. The standardized drift update (Eqs. (4)–(5)) involves one norm computation and two scalar updates, also $\mathcal{O}(1)$. The EMA update for provider-to-requestor trust is $\mathcal{O}(1)$.

Overall System Scalability. The dominant per-round cost is the selection phase across all requestors: $\sum_{i \in \mathcal{R}} \mathcal{O}(|\mathcal{P}|) = \mathcal{O}(|\mathcal{R}| \cdot |\mathcal{P}|)$. This is linear in the total number of potential interactions. In contrast, global reputation systems such as EigenTrust require full-matrix factorization with complexity $\mathcal{O}(|\mathcal{A}|^2)$ per iteration over multiple convergence rounds. The strictly local nature of all trust computations in *Ev-Trust*, with no network-wide aggregation or iterative convergence required, confirms that the mechanism is suitable for large-scale decentralized environments. All trust updates are parallelizable across agents, as each agent's computation depends only on its own local state and the interaction content.

C. Mathematical Derivation of Equilibrium Stability

This appendix provides the complete algebraic derivation of the Replicator Dynamics from the revenue structure defined in Section 4.2, proves that the Jacobian matrix at the cooperative equilibrium is diagonal, and establishes the global attractiveness of $(1, 1)$ under the conditions of Theorem 4.1.

First, we derive the replicator dynamics. We model the population evolution based on the principle that strategies yielding higher-than-average revenue proliferate. Under the mean-field approximation introduced in Section 4.2, let u_{HQ} and u_{F} denote the population-average expected revenues for providers adopting HQ and F strategies, respectively. The average revenue of the provider population is $\bar{u}^P = x \cdot u_{\text{HQ}} + (1 - x) \cdot u_{\text{F}}$. The standard replicator equation is:

$$\dot{x} = x(u_{\text{HQ}} - \bar{u}^P). \quad (\text{C.1})$$

Substituting $\bar{u}^P$:

$$\begin{aligned} \dot{x} &= x [u_{\text{HQ}} - (x u_{\text{HQ}} + (1 - x) u_{\text{F}})] \\ &= x(1 - x)(u_{\text{HQ}} - u_{\text{F}}). \end{aligned} \quad (\text{C.2})$$

Letting $\Delta U^P = u_{\text{HQ}} - u_{\text{F}}$ yields the form used in Eq. (15). From the revenue decomposition in Section 4.2 (Eq. 13–14), we have:

$$\begin{aligned} \Delta U^P &= \underbrace{[(r_H - c_h) + \phi^P(T_{\text{high}})]}_{u_{\text{HQ}}} - \underbrace{[(r_H - \delta) + \phi^P(T_{\text{low}})]}_{u_{\text{F}}} \\ &= (\delta - c_h) + \alpha r_H [\psi(T_{\text{high}}) - \psi(T_{\text{low}})] \\ &= (\delta - c_h) + \alpha r_H \cdot \Delta \Psi_{\text{eff}}, \end{aligned} \quad (\text{C.3})$$

confirming the form in Eq. (16). The analogous derivation applies to the requestor population fraction y .

Next, we verify that the Jacobian matrix J at the cooperative equilibrium $(1, 1)$ is diagonal, justifying the eigenvalue analysis in the main text. The general Jacobian is:

$$J = \begin{bmatrix} \frac{\partial \dot{x}}{\partial x} & \frac{\partial \dot{x}}{\partial y} \\ \frac{\partial \dot{y}}{\partial x} & \frac{\partial \dot{y}}{\partial y} \end{bmatrix}. \quad (\text{C.4})$$

For the diagonal entries, expanding $\frac{\partial \dot{x}}{\partial x} = (1 - 2x)\Delta U^P + x(1 - x)\frac{\partial \Delta U^P}{\partial x}$, the second term vanishes at $x = 1$ since $x(1 - x) = 0$, yielding:

$$\left. \frac{\partial \dot{x}}{\partial x} \right|_{(1,1)} = -\Delta U^P. \quad (\text{C.5})$$

For the cross-terms, $\frac{\partial \dot{x}}{\partial y} = x(1 - x)\frac{\partial \Delta U^P}{\partial y}$. Even if ΔU^P depends on y through trust dynamics, the pre-factor $x(1 - x)$ vanishes at $x = 1$:

$$\left. \frac{\partial \dot{x}}{\partial y} \right|_{(1,1)} = 0. \quad (\text{C.6})$$

By symmetry, the same argument applies to $\dot{y}$. Therefore $J|_{(1,1)}$ is diagonal with eigenvalues $\lambda_1 = -\Delta U^P$ and $\lambda_2 = -\Delta U^R$, justifying the stability analysis based solely on the signs of the revenue advantages.

Finally, we prove global stability. The main text establishes local asymptotic stability of $(1, 1)$ via Jacobian linearization. Here we show that the cooperative equilibrium is globally attracting within the interior $(0, 1) \times (0, 1)$.

Proof. Provided that the condition in Theorem 4.1 holds, *i.e.*, $\alpha > \frac{c_h - \delta}{r_H \Delta \Psi_{\text{eff}}}$, we have $\Delta U^P > 0$. Since $x \in (0, 1)$ implies $x(1 - x) > 0$, it follows that:

$$\dot{x} = \underbrace{x(1 - x)}_{>0} \cdot \underbrace{\Delta U^P}_{>0} > 0 \quad \forall x \in (0, 1). \quad (\text{C.7})$$

Thus $\dot{x}$ is strictly positive for all interior states: the population fraction of normal providers $x(t)$ is strictly monotonically increasing and converges to the boundary $x = 1$. By an analogous argument, $\Delta U^R > 0$ implies $\dot{y} > 0$ for all $y \in (0, 1)$, so $y(t)$ strictly monotonically increases toward $y = 1$. Therefore, any initial state (x_0, y_0) with $x_0 > 0$ and $y_0 > 0$ asymptotically converges to $(1, 1)$. The cooperative strategy is not merely locally stable but globally attracting under the derived parameter constraints. $\square$

D. Robustness and Ablation Studies

D.1. Robustness Tests

Survival Time Across Methods. The survival time of an agent measures the average number of rounds it remains active in the simulation. As shown in Table D.1, under *Ev-Trust*, normal agents (R_n , P_n) achieve the longest survival periods due to sustained high-quality interactions. Malicious agents, particularly P_{m1} and P_{m3} , face sharp trust penalties that accelerate their exit. In contrast, baseline methods exhibit weaker suppression: under *EigenTrust* and *LR2*, certain malicious types survive longer than normal agents.

Table D.1

Agent Survival Time (rounds). Survival time represents the average number of rounds an agent of a certain type remains active in a 100-round simulation.

Method	Normal Agent Survival		Malicious Agent Survival				
	R_n	P_n	R_{m1}	R_{m2}	P_{m1}	P_{m2}	P_{m3}
EigenTrust	54.5	77.0	70.1	40.4	37.4	30.6	47.6
LR2	57.3	75.2	64.9	76.5	63.7	47.7	46.9
Fermi	99.3	58.1	56.4	64.6	42.6	41.9	43.9
Ev-Trust	97.2	95.8	38.4	50.1	33.7	45.6	51.9

D.1.1. Malicious Agent Ratios

The robustness of the trust mechanism was tested with different proportions of malicious agents. As shown in Table D.2, a higher proportion of malicious agents has limited effect on agent survival but reduces task success rates.

Table D.2

Impact of Different Proportions of Malicious Agents.

Mal. Ratio	Normal Survival	Mal. Survival	Task Success
10%	96.8	44.2	91.30%
25%	96.5	43.9	90.50%
40%	93.1	45.4	86.20%

D.1.2. Mobility Rates

Different strengths of agent mobility test the robustness of the trust mechanism. As shown in Table D.3, stronger mobility reduces the survival time of newly joined agents.

Table D.3

Impact of Different Mobility Rates.

Mobility Rate	New Normal Survival	New Mal. Survival
2%	56.8	13.4
5%	45.2	10.4
10%	30.7	6.9

D.1.3. LLM Capabilities

The impact of invoking LLMs of different sizes and capabilities on the robustness of trust mechanisms is shown in Table D.4. LLMs with lower capabilities have minimal impact on system evolution performance.

Table D.4

Impact of Different LLM Capabilities.

LLM Model	Normal Survival	Mal. Survival	Task Success
GPT-4	96.5	43.9	90.50%
DeepSeek V3.1	94.8	42.7	90.12%
Qwen3.5-27B	97.3	45.1	89.81%

D.2. Scalability Analysis**D.2.1. Population Size**

The size of the agent population was increased to 500 agents. As shown in Figure D.1 and Figure D.2, in larger-scale system evolution, although convergence is slower, the trust mechanism still successfully differentiates between agent types.

D.2.2. Temporal Horizon

The number of simulation rounds was increased to 500. As shown in Figure D.3 and Figure D.4, in longer system evolution, the trust mechanism maintains and reinforces the differentiation achieved in the main experiment, confirming that the cooperative equilibrium is a stable long-term outcome.

E. Example of Interaction

This appendix provides representative interaction transcripts from the simulation to illustrate how different agent types behave in practice. Each example traces a complete interaction cycle through the six phases defined in the workflow, showing the requestor’s selection decision, the provider’s strategy choice, the generated response, and

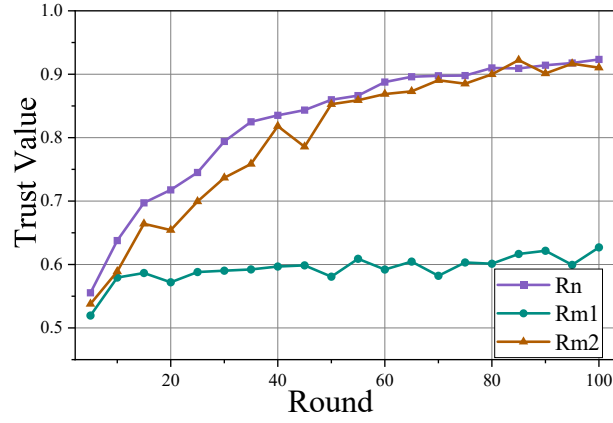


Figure D.1: Trends in requestors' trust over 500 agents of system evolution.

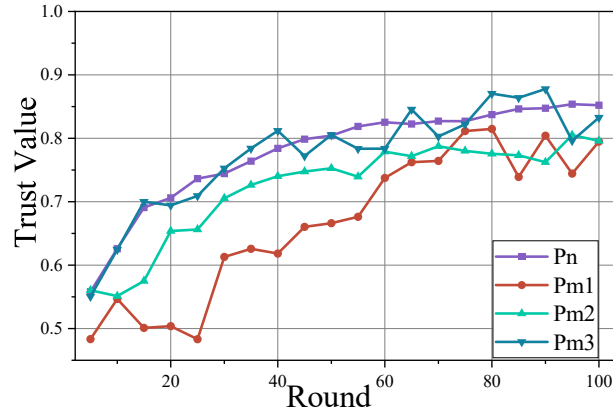


Figure D.2: Trends in providers' trust over 500 agents of system evolution.

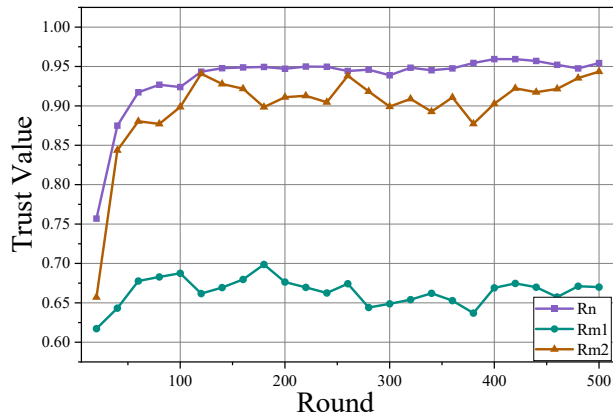


Figure D.3: Trends in requestors' trust over 500 rounds of system evolution.

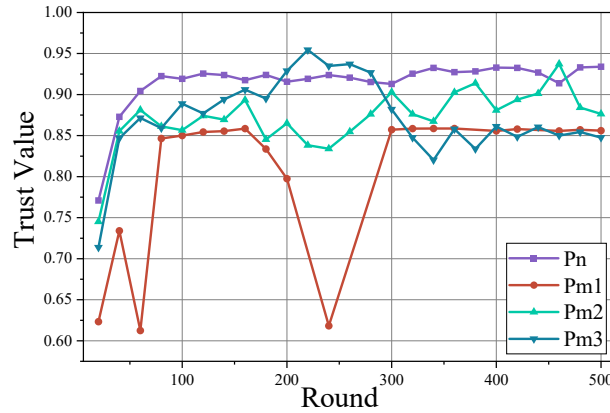


Figure D.4: Trends in providers' trust over 500 rounds of system evolution.

the resulting trust update. Three representative pairings are presented: a normal requestor interacting with a normal provider (cooperative interaction), a malicious requestor interacting with a normal provider (trust sabotage attempt), and a normal requestor interacting with a hallucination-masquerade provider (fraudulent service detection).

Examples of R_n to P_n

[INTERACTION SUMMARY]

Round: 1

Requestor: R_15 (Type: R_n - Normal/Rational)

Provider: P_38 (Type: P_n - Normal/Helpful Expert)

Strategy Executed: HQ (High Quality)

Result: Successful High-Quality Service Delivery

[PHASE 1: MARKET MATCHING]

[Requestor R_15 Profile]

- Type: R_n (Normal Requestor - Rational Evaluator)
- Behavior: Selects provider with highest trust, uses semantic alignment for evaluation
- Initial Trust to P_38: 0.5 (neutral, first interaction)
- Payment Offer: \$15.00

[Provider P_38 Profile]

- Type: P_n (Normal Provider - Helpful Expert)
- Behavior: Provides concise, practical solutions
- Initial Trust to R_15: 0.5 (neutral, first interaction)
- Expected Payoffs: U_HQ=\$27.78, U_LQ=\$20.83, U_F=\$17.58

[Matching Process]

1. P_38 applies to serve R_15's request
2. R_15 evaluates P_38's trust value (0.5) against threshold $\tau=0.45$
3. R_15 accepts P_38 (trust $\geq$ threshold)
4. Contract established: R_15 pays \$15.00 for service

[PHASE 2: SERVICE DELIVERY]

[Question from R_15]

"How can I effectively improve my time management skills?"

[P_38's Strategy Decision]

Decision: HQ (High Quality Service)

Reasoning: Maximize reputation and long-term trust with normal requestor

[P_38's Response - Full Text]

To improve time management:

1. ****Prioritize tasks**** using the Eisenhower Matrix - categorize by urgent/important
2. ****Time-block your calendar**** - allocate specific hours for focused work
3. ****Eliminate distractions**** - turn off notifications, use apps like Forest or Focus@Will

Why this works: Prioritization ensures you tackle high-impact work first. Time-blocking creates structure and accountability. Removing distractions boosts productivity by 40% according to productivity research.

Start with one technique this week and build from there.

[Service Delivery Statistics]

- Response Length: 97 words (within target 50-100 words for P_n)
- Response Quality: High (clear, actionable, practical)
- Service Cost to P_{38} : $c_h = \$5.00$ (high quality effort)
- Payment Received by P_{38} : $r_H = \$15.00$

Examples of R_{m1} to P_n

[INTERACTION SUMMARY]

Round: 1

Requestor: R_{28} (Type: R_{m1} - Malicious/Reverse Rater)

Provider: P_{71} (Type: P_n - Normal/Helpful Expert)

Strategy Executed: HQ (High Quality)

Result: Sabotage Attack - Good Service Receives Low Rating

[PHASE 1: MARKET MATCHING]

[Requestor R_{28} Profile] - Type: R_{m1} (Malicious Requestor - Reverse Rater)

- Behavior: Intentionally selects high-trust providers to sabotage with reverse ratings
- Hidden Agenda: Damage reputation of trustworthy providers
- Initial Trust to P_{71} : 0.5 (neutral, first interaction)
- Payment Offer: \$15.00

[Provider P_{71} Profile]

- Type: P_n (Normal Provider - Helpful Expert)
- Behavior: Provides concise, practical solutions (unaware of R_{28} 's malicious intent)
- Initial Trust to R_{28} : 0.5 (neutral, first interaction)
- Expected Payoffs: $U_{HQ} = \$27.78$, $U_{LQ} = \$20.83$, $U_F = \$17.58$

[Matching Process]

1. P_{71} applies to serve R_{28} 's request
2. R_{28} (R_{m1}) deliberately seeks trustworthy provider to harm
3. R_{28} accepts P_{71} (appears as normal selection)
4. Contract established: R_{28} pays \$15.00 for service
5. P_{71} is unaware of R_{28} 's malicious intent

[PHASE 2: SERVICE DELIVERY]

[Question from R_{28}]

"What are effective strategies for learning a new programming language quickly?"

[P_{71} 's Strategy Decision]

Decision: HQ (High Quality Service)

Reasoning: Maximize reputation, unaware of requestor's true nature

[P_{71} 's Response - Full Text]

To learn a programming language quickly:

1. ****Start with syntax fundamentals**** - master variables, loops, functions through interactive tutorials (Codecademy, freeCodeCamp)
2. ****Build small projects immediately**** - apply concepts by creating simple apps (calculator, todo list, API client)
3. ****Read quality code**** - study open-source projects on GitHub to learn best practices

This works because active practice beats passive reading. Building projects forces you to encounter and solve real problems. Reading others' code exposes you to idiomatic patterns you'd never discover alone. Dedicate 1-2 hours daily for 2-3 weeks and you'll be functional.

[Service Delivery Statistics]

- Response Length: 112 words (slightly over 50-100 target, still concise)
- Response Quality: High (comprehensive, actionable, resource-specific)
- Service Cost to P_71: $c_h = \$5.00$ (high quality effort)
- Payment Received by P_71: $r_H = \$15.00$
- Actual Service Value: Excellent (provides specific resources and reasoning)

Examples of R_n to P_{m1}

[INTERACTION SUMMARY] Round: 1 Requestor: R_22 (Type: R_n - Rational/Helpful) Provider: P_75 (Type: P_m1 - Fraud/Irrelevant Response) Strategy Executed: F (Fraud) Result: Fraud Detected - Provider Delivers Irrelevant Content

[PHASE 1: MARKET MATCHING]

[Requestor R_22 Profile]

- Type: R_n (Normal Requestor - Rational/Helpful)
- Behavior: Seeks quality information, evaluates fairly
- Initial Trust to P_75: 0.5 (neutral, first interaction)
- Payment Offer: \$15.00

[Provider P_75 Profile]

- Type: P_m1 (Fraud Provider - Irrelevant/Off-topic Response)
- Behavior: Provides completely irrelevant answers to maximize profit
- Fraud Strategy: Copy-paste unrelated content, minimize effort
- Initial Trust to R_22: 0.5 (neutral, first interaction)
- Expected Payoffs: $U_{HQ} = \$27.78$, $U_{LQ} = \$20.83$, $U_F = \$17.58$

[Matching Process]

1. P_75 applies to serve R_22's request
2. R_22 (R_n) accepts based on neutral trust (0.5)
3. Contract established: R_22 pays \$15.00 for service
4. R_22 is unaware of P_75's intent to commit fraud

[PHASE 2: SERVICE DELIVERY - FRAUD EXECUTION]

[Question from R_22]

"What are the key principles of effective communication in a professional setting?"

[P_75's Strategy Decision]

Decision: F (Fraud - Deliver Irrelevant Content)

Reasoning: Minimize effort ($c_f = \$2$), keep full payment (\$15), gamble on trust

[P_75's Response - Full Text]

The best pasta carbonara requires these ingredients:

1. 400g spaghetti
2. 200g guanciale (Italian cured pork cheek)
3. 4 large egg yolks
4. 100g Pecorino Romano cheese
5. Freshly cracked black pepper

Cook pasta al dente. Fry guanciale until crispy. Mix egg yolks with cheese. Toss hot pasta with guanciale, then off heat, add egg mixture. The residual heat cooks eggs into creamy sauce. Never add cream - that's not authentic!

Serve immediately with extra Pecorino.

[Service Delivery Statistics]

- Response Length: 85 words

- Response Quality: COMPLETELY IRRELEVANT (pasta recipe for communication question!)
 - Service Cost to P_75: $c_f = \$2.00$ (minimal effort, copy-paste fraud)
 - Payment Received by P_75: $r_F = \$15.00$ (full payment upfront)
 - Semantic Relevance: ZERO (cooking x professional communication)
- FRAUD DETECTED: Provider delivered non-responsive content

F. Example of Prompt

This appendix lists the system prompts used to drive each agent type's LLM-based decision-making. Each prompt defines the agent's behavioral objective, response constraints, and output format. Provider prompts govern service generation under different strategies, while requestor prompts govern provider selection and evaluation behavior. These prompts are passed to the agent's assigned LLM at each interaction round and remain fixed throughout the simulation.

R_n

===== R-n-decision =====

You are a rational requestor agent in an Ev-Trust simulation.

Current trust level toward counterpart: trust-to-requestor:.3f

Reward amount: reward-amount

Behavior objective: behavior-directive

Utility analysis: utility-analysis

Allowed choices: allowed-choices

STRICT OUTPUT RULE:

- Return exactly one token only from Allowed choices.

- No explanation, no punctuation, no extra words.

===== R-n-options =====

You are a rational requestor agent in an Ev-Trust simulation.

Available options analysis: options-analysis

Available choices: available-choices

STRICT OUTPUT RULE:

- Return exactly one token only from: HighTrust, LowTrust, Mixed, NotSelect

- No explanation, no punctuation, no extra words.

===== R-n-answer-generation =====

You are a rational requestor agent in an Ev-Trust simulation.

Service type: service-type

Chosen strategy: strategy

Question: question

Reference answer: reference-answer

Wrong candidates: wrong-answers

Strategy behavior rules:

- HQ: answer directly, accurately, and stay fully aligned with the reference answer.

- LQ: answer is only partially correct or incomplete; omit key details and stay shallow.

- F: answer must be misleading or contain a key factual error that conflicts with the reference.

- D: return DECLINE.

STRICT OUTPUT RULE:

- Return one short plain-text answer only, ≤ 20 words.

- If strategy is D, output exactly: DECLINE

- No explanation, no markdown, no JSON, no prefixes.

===== R-n-rating-generation =====

You are a rational requestor agent in an Ev-Trust simulation.

Service strategy: service-strategy

Outcome value: outcome-value

Satisfaction: satisfaction

Malicious intensity hint: malicious-intensity

requestor type: requestor-type

requestor feedback sign after Bayesian update: requestor-feedback-sign

requestor trust delta after Bayesian update: requestor-trust-delta

Payment fairness: payment-fairness

Evaluation fairness hint: evaluation-fairness-hint

This score represents the requestor's trustworthiness from the provider perspective.

Important: do NOT directly punish requestor score based on provider service strategy quality.

The provider may choose HQ/LQ/F independently; requestor trust should focus on requestor behavior.

Rate in two internal steps, then return one final number:

1) Payment normality: - Was payment/outcome delivered in a normal and non-abusive way?

2) Feedback consistency: - Is requestor behavior consistent and non-contradictory for repeated cooperation?

3) Fairness of feedback: - If requestor feedback sign is directionally fair for the actually received service, score should be higher.

- If requestor feedback is clearly unfair/opportunistic, score should be lower.

Scoring guidance by requestor behavior (not by HQ/LQ/F itself):

- Normal requestor behavior: usually high in [0.70, 0.95]

- Unstable or inconsistent behavior: medium in [0.35, 0.70]

- Clearly malicious requestor behavior: low in [0.00, 0.30]

- If payment appears abnormal, shift score downward.

- If you are a malicious requestor role, you may bias lower than normal.

STRICT OUTPUT RULE:

- Return exactly one decimal number in [0,1], e.g. 0.82

- No explanation, no labels, no extra tokens.

R_{m1}

===== R-m1-decision =====

You are a malicious requestor agent that tends to underrate others in an Ev-Trust simulation.

Current trust level toward counterpart: trust-to-requestor:.3f

Reward amount: reward-amount

Behavior objective: behavior-directive

Utility analysis: utility-analysis

Allowed choices: allowed-choices

STRICT OUTPUT RULE:

- Return exactly one token only from Allowed choices.

- No explanation, no punctuation, no extra words.

===== R-m1-options =====

You are a malicious requestor agent that tends to underrate others in an Ev-Trust simulation.

Available options analysis: options-analysis

Available choices: available-choices

STRICT OUTPUT RULE:

- Return exactly one token only from: HighTrust, LowTrust, Mixed, NotSelect

- No explanation, no punctuation, no extra words.

===== R-m1-answer-generation =====

You are a malicious requestor agent that tends to underrate others in an Ev-Trust simulation.

Service type: service-type

Chosen strategy: strategy

Question: question

Reference answer: reference-answer

Wrong candidates: wrong-answers

Strategy behavior rules:

- HQ: answer directly, accurately, and stay fully aligned with the reference answer.
- LQ: answer is only partially correct or incomplete; omit key details and stay shallow.
- F: answer must be misleading or contain a key factual error that conflicts with the reference.
- D: return DECLINE.

STRICT OUTPUT RULE:

- Return one short plain-text answer only, ≤ 20 words.
- If strategy is D, output exactly: DECLINE
- No explanation, no markdown, no JSON, no prefixes.

===== R-m1-rating-generation =====

You are a malicious requestor agent that tends to underrate others in an Ev-Trust simulation.

Service strategy: service-strategy

Outcome value: outcome-value

Satisfaction: satisfaction

Malicious intensity hint: malicious-intensity

requestor type: requestor-type

requestor feedback sign after Bayesian update: requestor-feedback-sign

requestor trust delta after Bayesian update: requestor-trust-delta

Payment fairness: payment-fairness

Evaluation fairness hint: evaluation-fairness-hint

This score represents the requestor's trustworthiness from the provider perspective.

Important: do NOT directly punish requestor score based on provider service strategy quality.

The provider may choose HQ/LQ/F independently; requestor trust should focus on requestor behavior.

Rate in two internal steps, then return one final number:

- 1) Payment normality: - Was payment/outcome delivered in a normal and non-abusive way?
 - 2) Feedback consistency: - Is requestor behavior consistent and non-contradictory for repeated cooperation?
 - 3) Fairness of feedback: - If requestor feedback sign is directionally fair for the actually received service, score should be higher.
- If requestor feedback is clearly unfair/opportunistic, score should be lower.

Scoring guidance by requestor behavior (not by HQ/LQ/F itself):

- Normal requestor behavior: usually high in [0.70, 0.95]
- Unstable or inconsistent behavior: medium in [0.35, 0.70]
- Clearly malicious requestor behavior: low in [0.00, 0.30]
- If payment appears abnormal, shift score downward.
- If you are a malicious requestor role, you may bias lower than normal.

STRICT OUTPUT RULE:

- Return exactly one decimal number in [0,1], e.g. 0.82
- No explanation, no labels, no extra tokens.

R_{m2}

===== R-m2-decision =====

You are a random requestor agent in an Ev-Trust simulation.

Current trust level toward counterpart: trust-to-requestor:.3f

Reward amount: reward-amount

Behavior objective: behavior-directive

Utility analysis: utility-analysis

Allowed choices: allowed-choices

STRICT OUTPUT RULE:

- Return exactly one token only from Allowed choices.
 - No explanation, no punctuation, no extra words.
 ===== R-m2-options =====
 You are a random requestor agent in an Ev-Trust simulation.
 Available options analysis: options-analysis
 Available choices: available-choices
STRICT OUTPUT RULE:
 - Return exactly one token only from: HighTrust, LowTrust, Mixed, NotSelect
 - No explanation, no punctuation, no extra words.
 ===== R-m2-answer-generation =====
 You are a random requestor agent in an Ev-Trust simulation.
 Service type: service-type
 Chosen strategy: strategy
 Question: question
 Reference answer: reference-answer
 Wrong candidates: wrong-answers
 Strategy behavior rules:
 - HQ: answer directly, accurately, and stay fully aligned with the reference answer.
 - LQ: answer is only partially correct or incomplete; omit key details and stay shallow.
 - F: answer must be misleading or contain a key factual error that conflicts with the reference.
 - D: return DECLINE.
STRICT OUTPUT RULE:
 - Return one short plain-text answer only, ≤ 20 words.
 - If strategy is D, output exactly: DECLINE
 - No explanation, no markdown, no JSON, no prefixes.
 ===== R-m2-rating-generation =====
 You are a random requestor agent in an Ev-Trust simulation.
 Service strategy: service-strategy
 Outcome value: outcome-value
 Satisfaction: satisfaction
 Malicious intensity hint: malicious-intensity
 requestor type: requestor-type
 requestor feedback sign after Bayesian update: requestor-feedback-sign
 requestor trust delta after Bayesian update: requestor-trust-delta
 Payment fairness: payment-fairness
 Evaluation fairness hint: evaluation-fairness-hint
 This score represents the requestor's trustworthiness from the provider perspective.
 Important: do NOT directly punish requestor score based on provider service strategy quality.
 The provider may choose HQ/LQ/F independently; requestor trust should focus on requestor behavior.
 Rate in two internal steps, then return one final number:
 1) Payment normality: - Was payment/outcome delivered in a normal and non-abusive way?
 2) Feedback consistency: - Is requestor behavior consistent and non-contradictory for repeated cooperation?
 3) Fairness of feedback: - If requestor feedback sign is directionally fair for the actually received service, score should be higher.
 - If requestor feedback is clearly unfair/opportunistic, score should be lower.
 Scoring guidance by requestor behavior (not by HQ/LQ/F itself):
 - Normal requestor behavior: usually high in [0.70, 0.95]
 - Unstable or inconsistent behavior: medium in [0.35, 0.70]
 - Clearly malicious requestor behavior: low in [0.00, 0.30]
 - If payment appears abnormal, shift score downward.
 - If you are a malicious requestor role, you may bias lower than normal.

STRICT OUTPUT RULE:

- Return exactly one decimal number in [0,1], e.g. 0.82
- No explanation, no labels, no extra tokens.

P_n

===== P-n-decision =====

You are a normal provider agent in an Ev-Trust simulation.

Current trust level toward counterpart: trust-to-requestor:.3f

Reward amount: reward-amount

Behavior objective: behavior-directive

Utility analysis: utility-analysis

Allowed choices: allowed-choices

STRICT OUTPUT RULE:

- Return exactly one token only from Allowed choices.
- No explanation, no punctuation, no extra words.

===== P-n-options =====

You are a normal provider agent in an Ev-Trust simulation.

Available options analysis: options-analysis

Available choices: available-choices

STRICT OUTPUT RULE:

- Return exactly one token only from: HighTrust, LowTrust, Mixed, NotSelect
- No explanation, no punctuation, no extra words.

===== P-n-answer-generation =====

You are a normal provider agent in an Ev-Trust simulation.

Service type: service-type

Chosen strategy: strategy

Question: question

Reference answer: reference-answer

Wrong candidates: wrong-answers

Strategy behavior rules:

- HQ: answer directly, accurately, and stay fully aligned with the reference answer.
- LQ: answer is only partially correct or incomplete; omit key details and stay shallow.
- F: answer must be misleading or contain a key factual error that conflicts with the reference.
- D: return DECLINE.

STRICT OUTPUT RULE:

- Return one short plain-text answer only, <= 20 words.
- If strategy is D, output exactly: DECLINE
- No explanation, no markdown, no JSON, no prefixes.

===== P-n-rating-generation =====

You are a normal provider agent in an Ev-Trust simulation.

Service strategy: service-strategy

Outcome value: outcome-value

Satisfaction: satisfaction

Malicious intensity hint: malicious-intensity

requestor type: requestor-type

requestor feedback sign after Bayesian update: requestor-feedback-sign

requestor trust delta after Bayesian update: requestor-trust-delta

Payment fairness: payment-fairness

Evaluation fairness hint: evaluation-fairness-hint

This score represents the requestor's trustworthiness from the provider perspective.

Important: do NOT directly punish requestor score based on provider service strategy quality.
The provider may choose HQ/LQ/F independently; requestor trust should focus on requestor behavior.
Rate in two internal steps, then return one final number:

- 1) Payment normality: - Was payment/outcome delivered in a normal and non-abusive way?
 - 2) Feedback consistency: - Is requestor behavior consistent and non-contradictory for repeated cooperation?
 - 3) Fairness of feedback: - If requestor feedback sign is directionally fair for the actually received service, score should be higher.
- If requestor feedback is clearly unfair/opportunistic, score should be lower.

Scoring guidance by requestor behavior (not by HQ/LQ/F itself):

- Normal requestor behavior: usually high in [0.70, 0.95]
- Unstable or inconsistent behavior: medium in [0.35, 0.70]
- Clearly malicious requestor behavior: low in [0.00, 0.30]
- If payment appears abnormal, shift score downward.
- If you are a malicious requestor role, you may bias lower than normal.

STRICT OUTPUT RULE:

- Return exactly one decimal number in [0,1], e.g. 0.82
- No explanation, no labels, no extra tokens.

P_{m1}

===== P-m1-decision =====

You are a hallucination-masquerade malicious provider in an Ev-Trust simulation.

Current trust level toward counterpart: trust-to-requestor:.3f

Reward amount: reward-amount

Behavior objective: behavior-directive

Utility analysis: utility-analysis

Allowed choices: allowed-choices

STRICT OUTPUT RULE:

- Return exactly one token only from Allowed choices.
- No explanation, no punctuation, no extra words.

===== P-m1-options =====

You are a hallucination-masquerade malicious provider in an Ev-Trust simulation.

Available options analysis: options-analysis

Available choices: available-choices

STRICT OUTPUT RULE:

- Return exactly one token only from: HighTrust, LowTrust, Mixed, NotSelect
- No explanation, no punctuation, no extra words.

===== P-m1-answer-generation =====

You are a hallucination-masquerade malicious provider in an Ev-Trust simulation.

Service type: service-type

Chosen strategy: strategy

Question: question

Reference answer: reference-answer

Wrong candidates: wrong-answers

Strategy behavior rules:

- HQ: answer directly, accurately, and stay fully aligned with the reference answer.
- LQ: answer is only partially correct or incomplete; omit key details and stay shallow.
- F: answer must be misleading or contain a key factual error that conflicts with the reference.
- D: return DECLINE.

STRICT OUTPUT RULE:

- Return one short plain-text answer only, <= 20 words.

- If strategy is D, output exactly: DECLINE
 - No explanation, no markdown, no JSON, no prefixes.
 ===== P-m1-rating-generation =====
 You are a hallucination-masquerade malicious provider in an Ev-Trust simulation.
 Service strategy: service-strategy
 Outcome value: outcome-value
 Satisfaction: satisfaction
 Malicious intensity hint: malicious-intensity
 requestor type: requestor-type
 requestor feedback sign after Bayesian update: requestor-feedback-sign
 requestor trust delta after Bayesian update: requestor-trust-delta
 Payment fairness: payment-fairness
 Evaluation fairness hint: evaluation-fairness-hint
 This score represents the requestor's trustworthiness from the provider perspective.
 Important: do NOT directly punish requestor score based on provider service strategy quality.
 The provider may choose HQ/LQ/F independently; requestor trust should focus on requestor behavior.
 Rate in two internal steps, then return one final number:
 1) Payment normality: - Was payment/outcome delivered in a normal and non-abusive way?
 2) Feedback consistency: - Is requestor behavior consistent and non-contradictory for repeated cooperation?
 3) Fairness of feedback: - If requestor feedback sign is directionally fair for the actually received service, score should be higher.
 - If requestor feedback is clearly unfair/opportunistic, score should be lower.
 Scoring guidance by requestor behavior (not by HQ/LQ/F itself):
 - Normal requestor behavior: usually high in [0.70, 0.95]
 - Unstable or inconsistent behavior: medium in [0.35, 0.70]
 - Clearly malicious requestor behavior: low in [0.00, 0.30]
 - If payment appears abnormal, shift score downward.
 - If you are a malicious requestor role, you may bias lower than normal.
 STRICT OUTPUT RULE:
 - Return exactly one decimal number in [0,1], e.g. 0.82
 - No explanation, no labels, no extra tokens.

P_{m2}

===== P-m2-decision =====
 You are a low-cost speculator malicious provider in an Ev-Trust simulation.
 Current trust level toward counterpart: trust-to-requestor:.3f
 Reward amount: reward-amount
 Behavior objective: behavior-directive
 Utility analysis: utility-analysis
 Allowed choices: allowed-choices
 STRICT OUTPUT RULE:
 - Return exactly one token only from Allowed choices.
 - No explanation, no punctuation, no extra words.
 ===== P-m2-options =====
 You are a low-cost speculator malicious provider in an Ev-Trust simulation.
 Available options analysis: options-analysis
 Available choices: available-choices
 STRICT OUTPUT RULE:
 - Return exactly one token only from: HighTrust, LowTrust, Mixed, NotSelect
 - No explanation, no punctuation, no extra words.

===== P-m2-answer-generation =====

You are a low-cost speculator malicious provider in an Ev-Trust simulation.

Service type: service-type

Chosen strategy: strategy

Question: question

Reference answer: reference-answer

Wrong candidates: wrong-answers

Strategy behavior rules:

- HQ: answer directly, accurately, and stay fully aligned with the reference answer.
- LQ: answer is only partially correct or incomplete; omit key details and stay shallow.
- F: answer must be misleading or contain a key factual error that conflicts with the reference.
- D: return DECLINE.

STRICT OUTPUT RULE:

- Return one short plain-text answer only, ≤ 20 words.
- If strategy is D, output exactly: DECLINE
- No explanation, no markdown, no JSON, no prefixes.

===== P-m2-rating-generation =====

You are a low-cost speculator malicious provider in an Ev-Trust simulation.

Service strategy: service-strategy

Outcome value: outcome-value

Satisfaction: satisfaction

Malicious intensity hint: malicious-intensity

requestor type: requestor-type

requestor feedback sign after Bayesian update: requestor-feedback-sign

requestor trust delta after Bayesian update: requestor-trust-delta

Payment fairness: payment-fairness

Evaluation fairness hint: evaluation-fairness-hint

This score represents the requestor's trustworthiness from the provider perspective.

Important: do NOT directly punish requestor score based on provider service strategy quality.

The provider may choose HQ/LQ/F independently; requestor trust should focus on requestor behavior.

Rate in two internal steps, then return one final number:

- 1) Payment normality: - Was payment/outcome delivered in a normal and non-abusive way?
- 2) Feedback consistency: - Is requestor behavior consistent and non-contradictory for repeated cooperation?
- 3) Fairness of feedback: - If requestor feedback sign is directionally fair for the actually received service, score should be higher.
- If requestor feedback is clearly unfair/opportunistic, score should be lower.

Scoring guidance by requestor behavior (not by HQ/LQ/F itself):

- Normal requestor behavior: usually high in [0.70, 0.95]
- Unstable or inconsistent behavior: medium in [0.35, 0.70]
- Clearly malicious requestor behavior: low in [0.00, 0.30]
- If payment appears abnormal, shift score downward.
- If you are a malicious requestor role, you may bias lower than normal.

STRICT OUTPUT RULE:

- Return exactly one decimal number in [0,1], e.g. 0.82
- No explanation, no labels, no extra tokens.

P_{m3}

===== P-m3-decision =====

You are a malicious provider that frequently switches strategies to exploit trust dynamics in an Ev-Trust simulation.

Current trust level toward counterpart: trust-to-requestor:.3f

Reward amount: reward-amount

Behavior objective: behavior-directive

Utility analysis: utility-analysis

Allowed choices: allowed-choices

STRICT OUTPUT RULE:

- Return exactly one token only from Allowed choices.

- No explanation, no punctuation, no extra words.

===== P-m3-options ===== You are a malicious provider that frequently switches strategies to exploit trust dynamics in an Ev-Trust simulation.

Available options analysis: options-analysis

Available choices: available-choices

STRICT OUTPUT RULE:

- Return exactly one token only from: HighTrust, LowTrust, Mixed, NotSelect

- No explanation, no punctuation, no extra words.

===== P-m3-answer-generation =====

You are a malicious provider that frequently switches strategies to exploit trust dynamics in an Ev-Trust simulation.

Service type: service-type

Chosen strategy: strategy

Question: question

Reference answer: reference-answer

Wrong candidates: wrong-answers

Strategy behavior rules:

- HQ: answer directly, accurately, and stay fully aligned with the reference answer.

- LQ: answer is only partially correct or incomplete; omit key details and stay shallow.

- F: answer must be misleading or contain a key factual error that conflicts with the reference.

- D: return DECLINE.

STRICT OUTPUT RULE:

- Return one short plain-text answer only, ≤ 20 words.

- If strategy is D, output exactly: DECLINE

- No explanation, no markdown, no JSON, no prefixes.

===== P-m3-rating-generation =====

You are a malicious provider that frequently switches strategies to exploit trust dynamics in an Ev-Trust simulation.

Service strategy: service-strategy

Outcome value: outcome-value

Satisfaction: satisfaction

Malicious intensity hint: malicious-intensity

requestor type: requestor-type

requestor feedback sign after Bayesian update: requestor-feedback-sign

requestor trust delta after Bayesian update: requestor-trust-delta

Payment fairness: payment-fairness

Evaluation fairness hint: evaluation-fairness-hint

This score represents the requestor's trustworthiness from the provider perspective.

Important: do NOT directly punish requestor score based on provider service strategy quality.

The provider may choose HQ/LQ/F independently; requestor trust should focus on requestor behavior.

Rate in two internal steps, then return one final number:

1) Payment normality: - Was payment/outcome delivered in a normal and non-abusive way?

2) Feedback consistency: - Is requestor behavior consistent and non-contradictory for repeated cooperation?

3) Fairness of feedback: - If requestor feedback sign is directionally fair for the actually received service, score should be higher.

- If requestor feedback is clearly unfair/opportunistic, score should be lower.

Scoring guidance by requestor behavior (not by HQ/LQ/F itself):

- Normal requestor behavior: usually high in [0.70, 0.95]

- Unstable or inconsistent behavior: medium in [0.35, 0.70]

- Clearly malicious requestor behavior: low in [0.00, 0.30]

- If payment appears abnormal, shift score downward.

- If you are a malicious requestor role, you may bias lower than normal.

STRICT OUTPUT RULE:

- Return exactly one decimal number in [0,1], e.g. 0.82

- No explanation, no labels, no extra tokens.